Dynasty Prospect Model

College-to-NFL Regression Analysis

Target: b2s_score | Draft classes: 2011-2022 | Positions: WR, RB, TE

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METHODOLOGY

Training data : 2011-2022 NFL draft classes (WR/RB/TE)
College data  : CFBD API — season totals, usage, PPA, recruiting
NFL outcomes  : nflverse player_stats (PPR fantasy scoring)
Pre-draft data: nflmockdraftdatabase.com consensus board (2016+)

Target — B2S Score:
  WR/RB: avg PPR PPG of best-2 seasons, first 3 NFL years (>=8 games)
  TE:    best single-season PPG, first 3 NFL years (>=8 games)

Analysis includes:
  * Univariate R2 scan — pairwise complete, 500-rep bootstrap CI
  * Multivariate OLS — standardized features (coefficients comparable)
  * Lasso / Ridge regularized regression with CV alpha selection
  * Leave-one-year-out cross-validation (temporal honesty)
  * Era sensitivity analysis (2011-15 / 2016-19 / 2020-22 / All)
  * Derived/interaction feature engineering (15 new features)
  * JJ Zachariason benchmark — his metric claims vs our data
  * Missing data inventory and improvement roadmap
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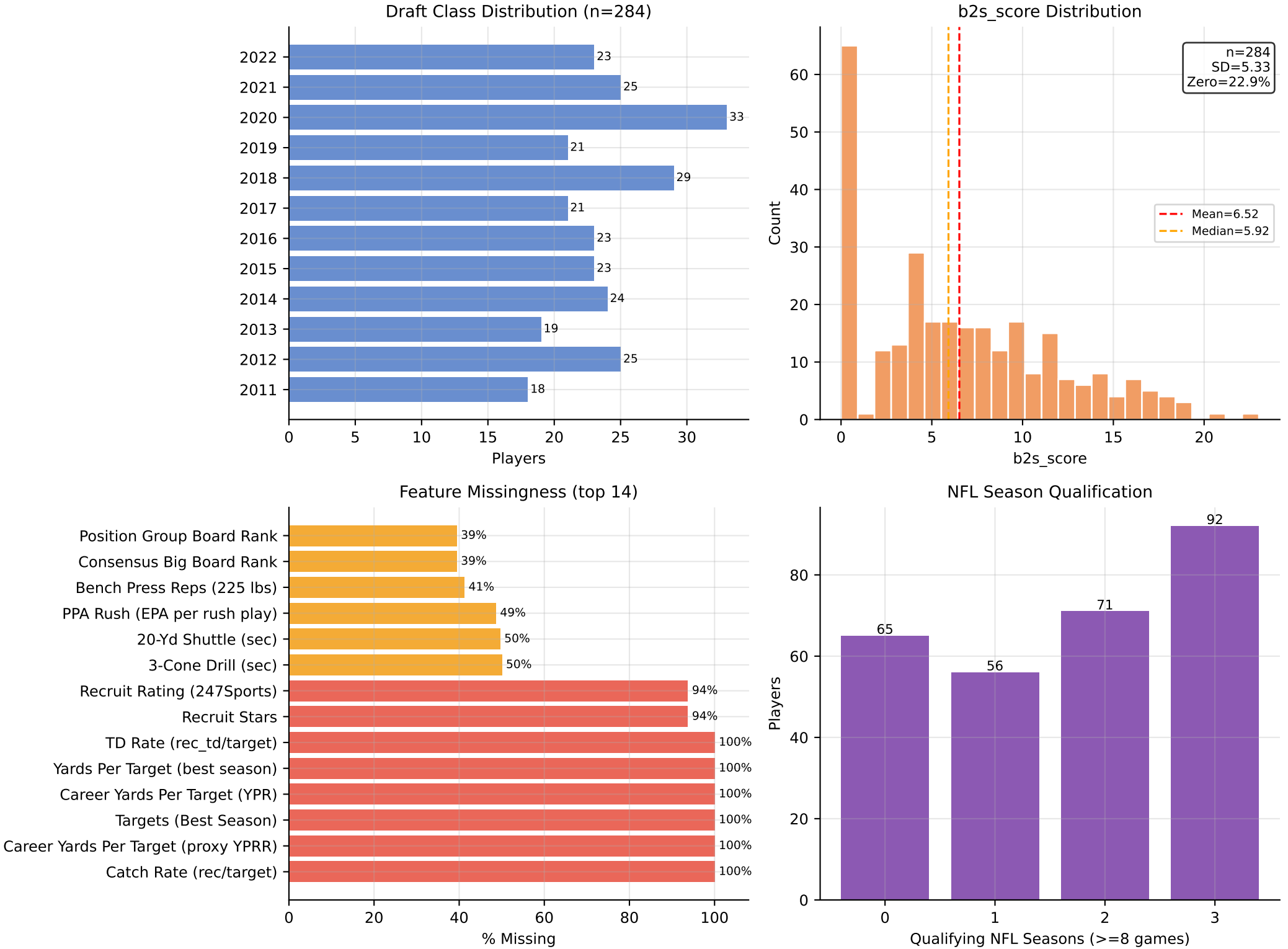
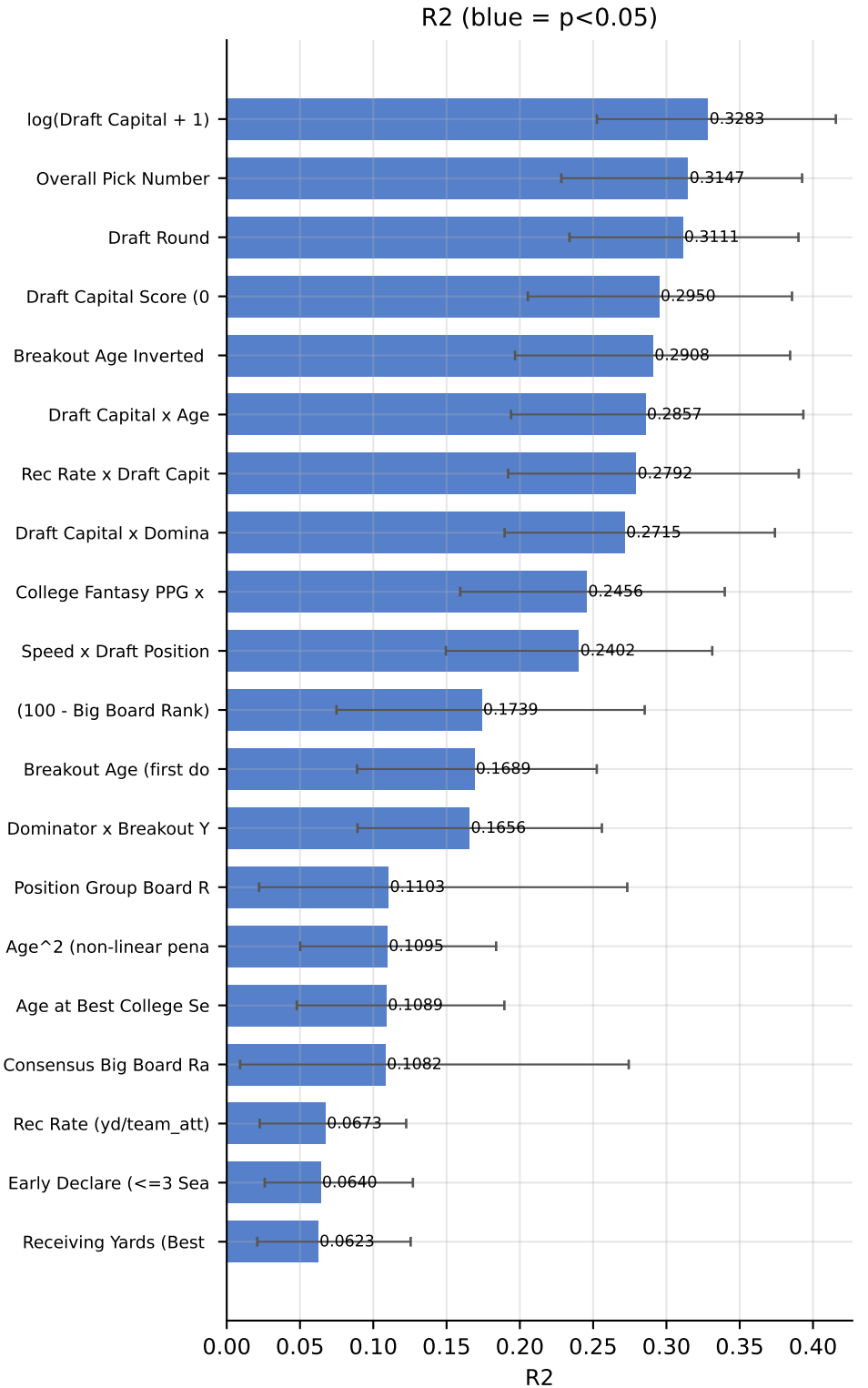
WR — Data Overview



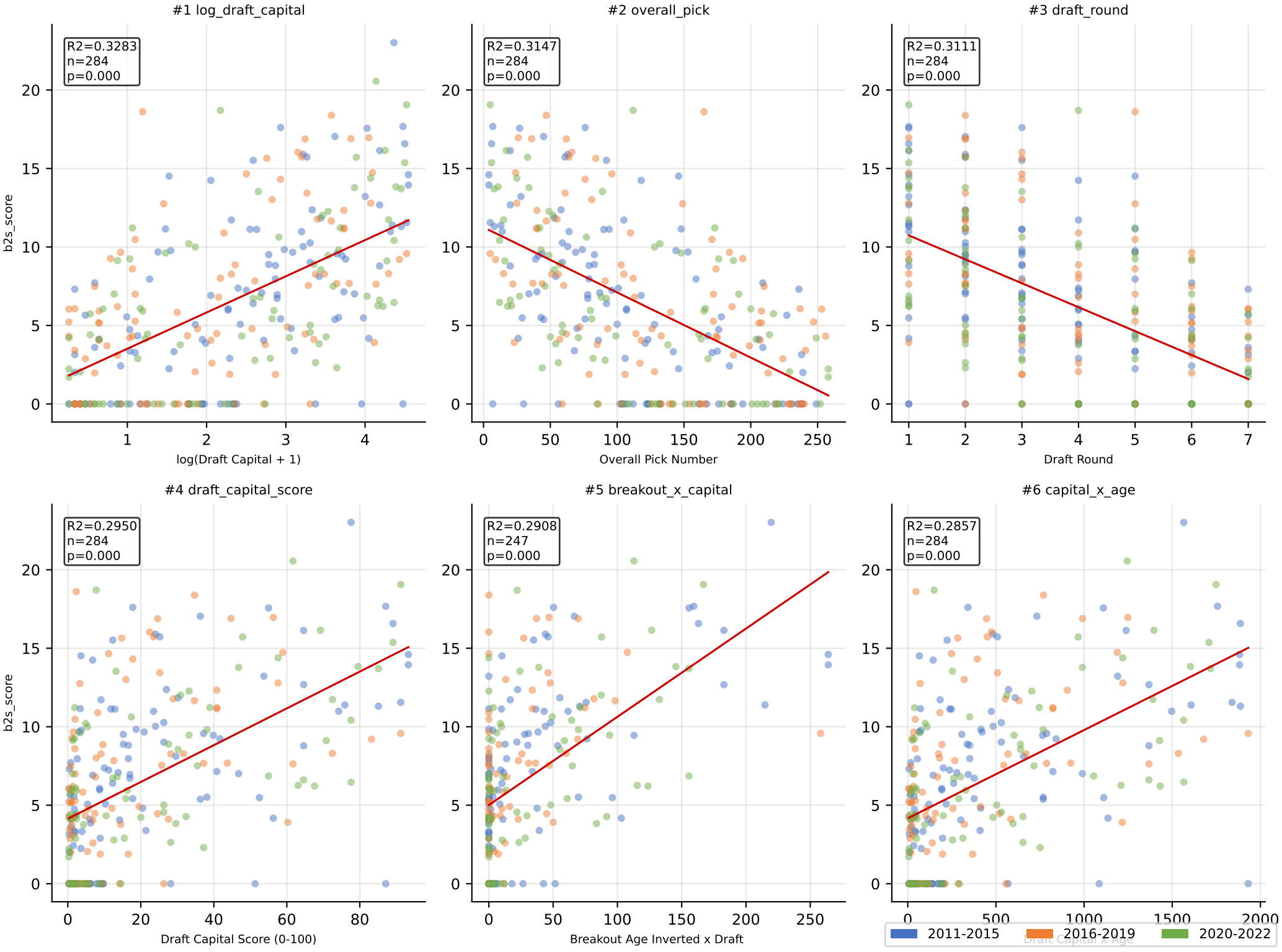
WR — Univariate R2 Ranking vs b2s_score

Rank	Feature	N	R2	95% CI	r	p-val
1	log(Draft Capital +	284	0.3283	[0.253, 0.416]	+0.573	0.000
2	Overall Pick Numb	284	0.3147	[0.228, 0.393]	-0.561	0.000
3	Draft Round	284	0.3111	[0.234, 0.390]	-0.558	0.000
4	Draft Capital Score (0	284	0.2950	[0.205, 0.386]	+0.543	0.000
5	Breakout Age Inverted	247	0.2908	[0.197, 0.384]	+0.539	0.000
6	Draft Capital x Ag	284	0.2857	[0.194, 0.393]	+0.534	0.000
7	Rec Rate x Draft Ca	284	0.2792	[0.192, 0.390]	+0.528	0.000
8	Draft Capital x Domi	284	0.2715	[0.190, 0.374]	+0.521	0.000
9	College Fantasy PPG x	284	0.2456	[0.159, 0.340]	+0.496	0.000
10	Speed x Draft Position (	284	0.2402	[0.150, 0.331]	+0.490	0.000
11	(100 - Big Board Rank	172	0.1739	[0.075, 0.285]	+0.417	0.000
12	Breakout Age (first dom	247	0.1689	[0.089, 0.252]	-0.411	0.000
13	Dominator x Breakout	247	0.1656	[0.089, 0.256]	+0.407	0.000
14	Position Group Board	172	0.1103	[0.022, 0.273]	-0.332	0.000
15	Age^2 (non-linear pe	284	0.1095	[0.050, 0.184]	-0.331	0.000
16	Age at Best College S	284	0.1089	[0.048, 0.189]	-0.330	0.000
17	Consensus Big Board	172	0.1082	[0.009, 0.274]	-0.329	0.000
18	Rec Rate (yd/team_att)	284	0.0673	[0.023, 0.122]	+0.259	0.000
19	Early Declare (<=3 Se	284	0.0640	[0.026, 0.127]	+0.253	0.000
20	Receiving Yards (Best S	282	0.0623	[0.021, 0.126]	+0.250	0.000

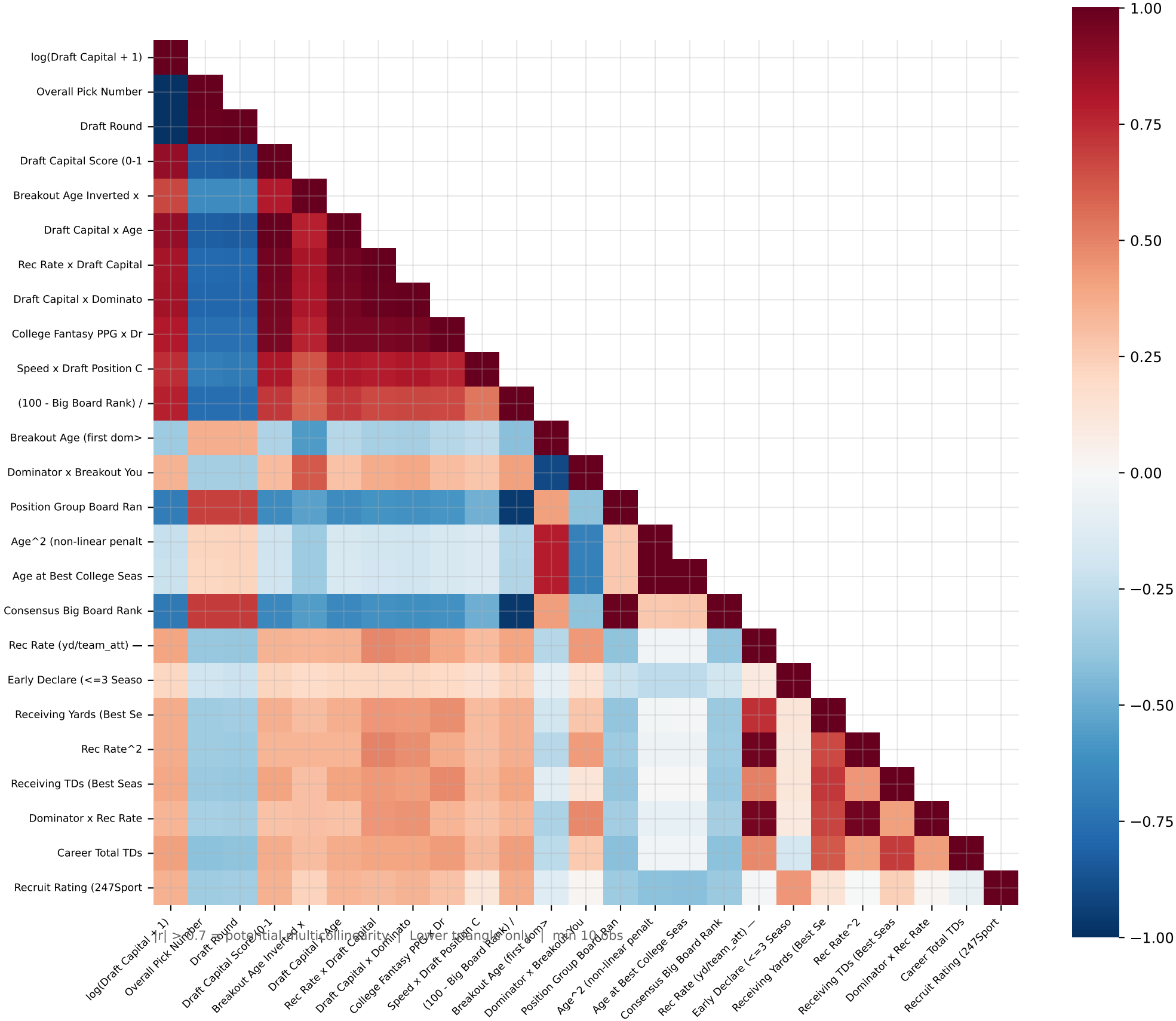


Green = R2>=0.05 & p<0.05 | Yellow = R2>=0.02 | 95% CI from 500-rep bootstrap

WR — Top Feature Scatter Plots vs b2s_score



WR — Feature Correlation Heatmap (Pearson r)

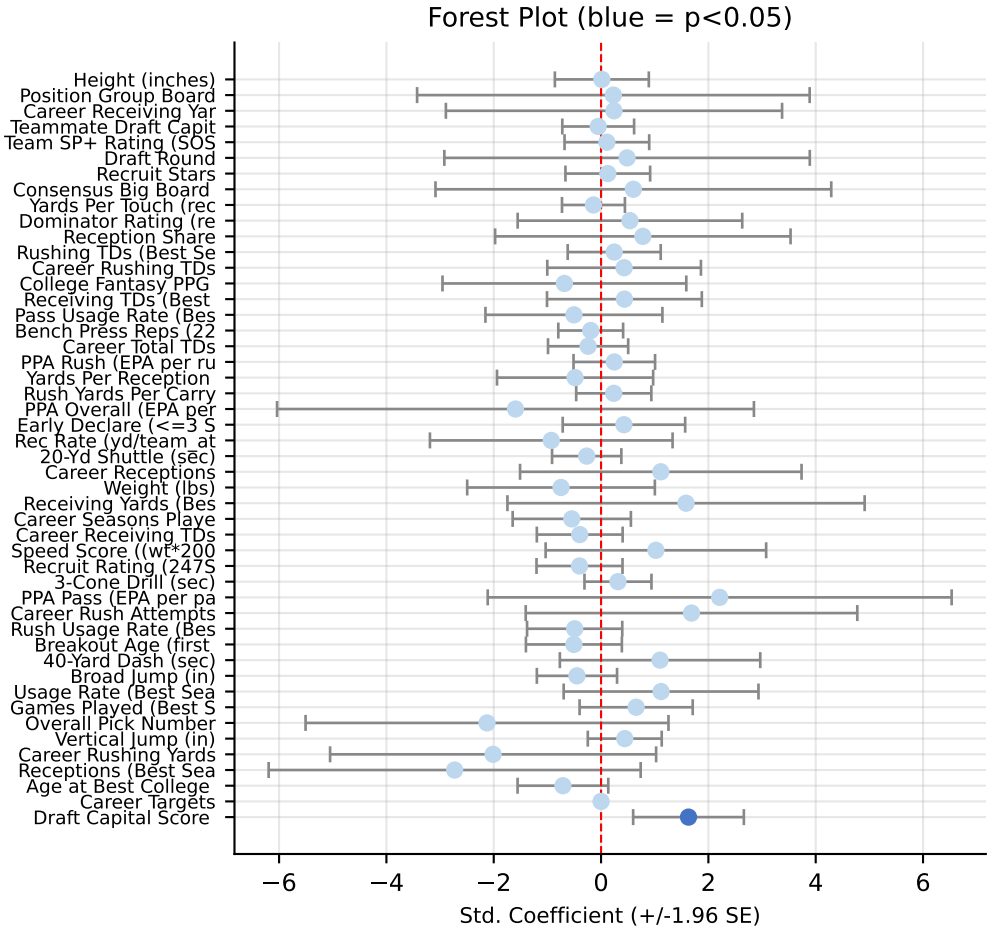


WR — OLS Regression vs b2s_score

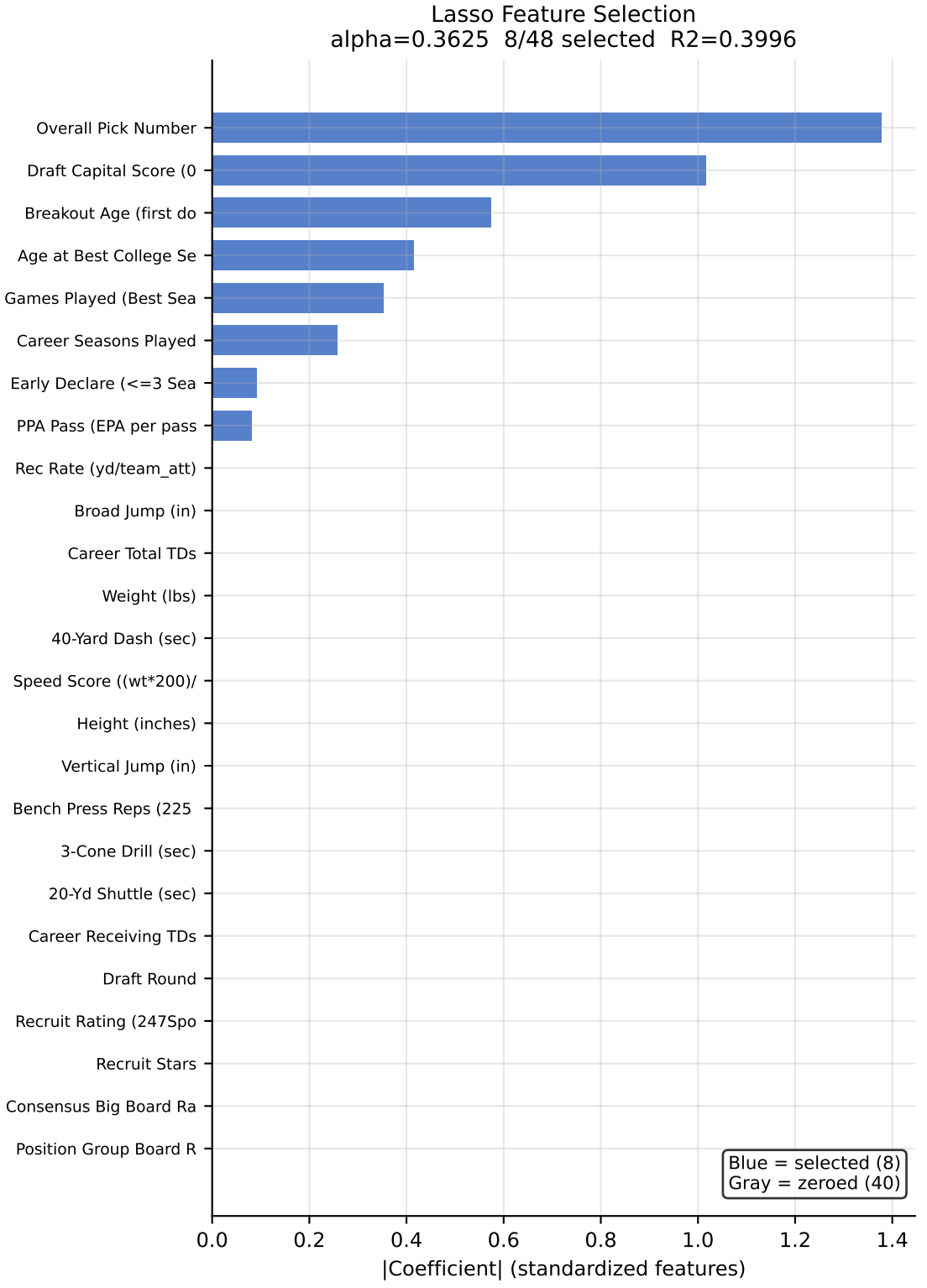
R2=0.4689 Adj-R2=0.3659 F=4.55 (p=0.0000) n=284 [Features standardized for comparability]

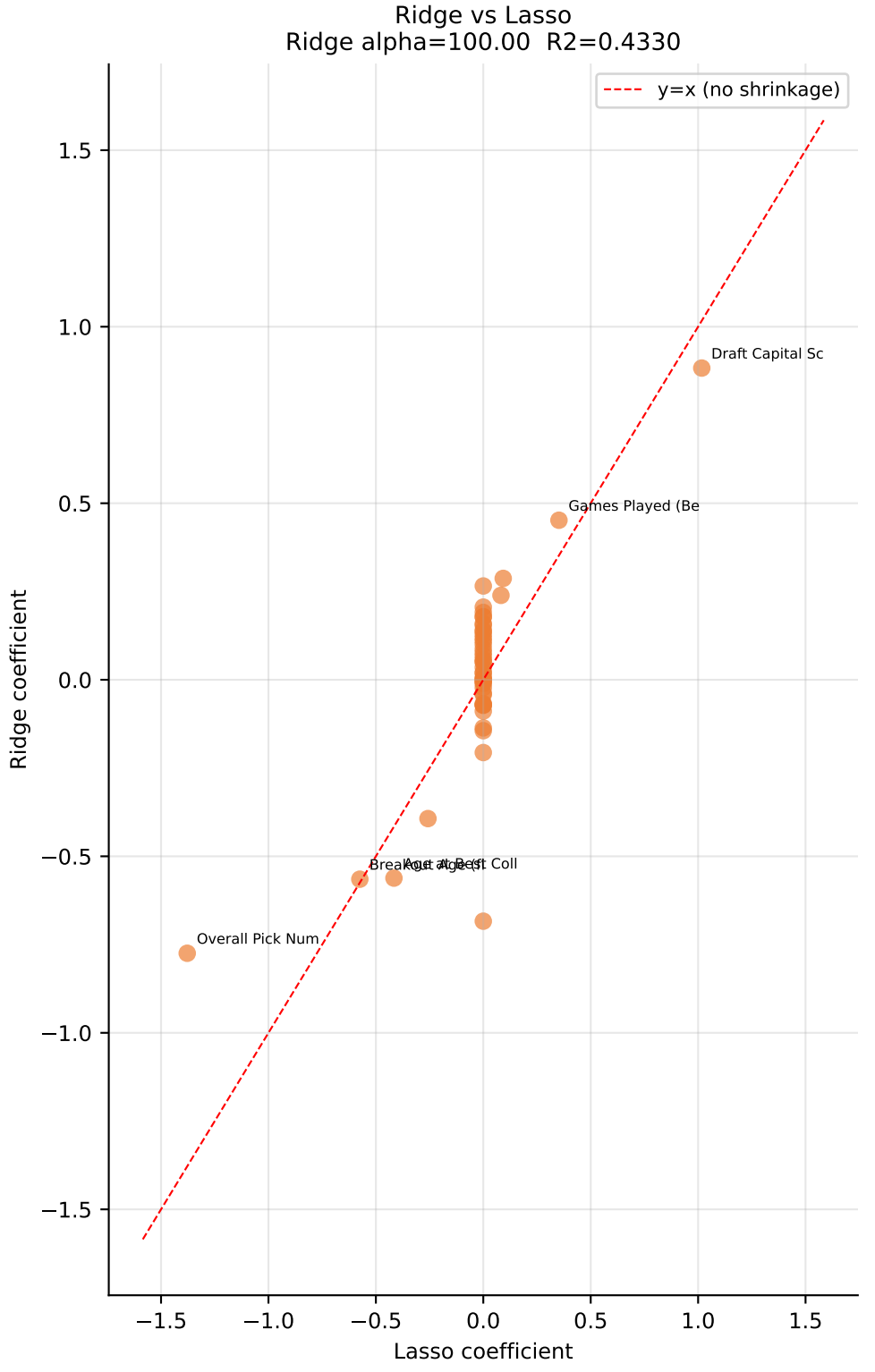
Feature	Coef	SE	t	p-val	VIF
aft Capital Score (0	+1.6298	0.5262	+3.10	0.002	4.4
Career Targets	+0.0000	0.0000	+1.76	0.079	--
e at Best College Se	-0.7104	0.4311	-1.65	0.101	2.9
eceptions (Best Sea	-2.7283	1.7686	-1.54	0.124	49.3
Career Rushing Yar	-2.0116	1.5499	-1.30	0.196	37.9
Vertical Jump (in)	+0.4417	0.3514	+1.26	0.210	1.9
Overall Pick Numbe	-2.1249	1.7253	-1.23	0.219	47.0
mes Played (Best Se	+0.6540	0.5377	+1.22	0.225	4.6
sage Rate (Best Sea	+1.1184	0.9261	+1.21	0.228	13.5
Broad Jump (in)	-0.4491	0.3813	-1.18	0.240	2.3
40-Yard Dash (sec	+1.0994	0.9527	+1.15	0.250	14.3
akout Age (first dor	-0.5063	0.4562	-1.11	0.268	3.3
sh Usage Rate (Best	-0.4920	0.4525	-1.09	0.278	3.2
Career Rush Attemp	+1.6855	1.5762	+1.07	0.286	39.2
A Pass (EPA per pas	+2.2106	2.2054	+1.00	0.317	76.7
3-Cone Drill (sec)	+0.3152	0.3184	+0.99	0.323	1.6
cruit Rating (247Sp	-0.4015	0.4089	-0.98	0.327	2.6
ed Score ((wt*200)/	+1.0218	1.0483	+0.97	0.331	17.3
Career Receiving TI	-0.3974	0.4079	-0.97	0.331	inf
Career Seasons Play	-0.5457	0.5620	-0.97	0.333	5.0
ceiving Yards (Best	+1.5835	1.6982	+0.93	0.352	45.5
Weight (lbs)	-0.7460	0.8921	-0.84	0.404	12.6
Career Receptions	+1.1134	1.3382	+0.83	0.406	28.2
20-Yd Shuttle (sec	-0.2674	0.3294	-0.81	0.418	1.7
c Rate (yd/team_att	-0.9280	1.1536	-0.80	0.422	21.0
ly Declare (<=3 Se	+0.4264	0.5821	+0.73	0.465	5.3
A Overall (EPA per p	-1.5944	2.2669	-0.70	0.483	81.1
sh Yards Per Carry (	+0.2368	0.3573	+0.66	0.508	2.0
rds Per Reception (R	-0.4841	0.7423	-0.65	0.515	8.7
A Rush (EPA per rus	+0.2467	0.3876	+0.64	0.525	2.4
Career Total TDs	-0.2415	0.3811	-0.63	0.527	inf
nch Press Reps (225	-0.1927	0.3081	-0.63	0.532	1.5
ss Usage Rate (Best	-0.5055	0.8412	-0.60	0.548	11.2
ceiving TDs (Best Se	+0.4354	0.7360	+0.59	0.555	8.5
llege Fantasy PPG (R	-0.6831	1.1589	-0.59	0.556	21.2
Career Rushing TD	+0.4296	0.7304	+0.59	0.557	inf
shing TDs (Best Sea	+0.2452	0.4421	+0.55	0.580	3.1
Reception Share	+0.7786	1.4045	+0.55	0.580	31.1
minator Rating (rec_	+0.5383	1.0676	+0.50	0.615	18.0
rds Per Touch (rec+r	-0.1416	0.2991	-0.47	0.636	1.4
nsensus Big Board (	+0.6022	1.8814	+0.32	0.749	55.8
Recruit Stars	+0.1255	0.4027	+0.31	0.756	2.6
Draft Round	+0.4847	1.7373	+0.28	0.781	47.6
eam SP+ Rating (SO	+0.1083	0.4025	+0.27	0.788	2.6
eammate Draft Cap	-0.0532	0.3413	-0.16	0.876	1.8
Career Receiving Ya	+0.2408	1.5983	+0.15	0.880	40.3
sition Group Board (	+0.2298	1.8661	+0.12	0.902	54.9
Height (inches)	+0.0146	0.4467	+0.03	0.974	3.1

Green=p<0.05 Yellow=p<0.10 Red VIF=>10



WR — Regularized Regression vs b2s_score

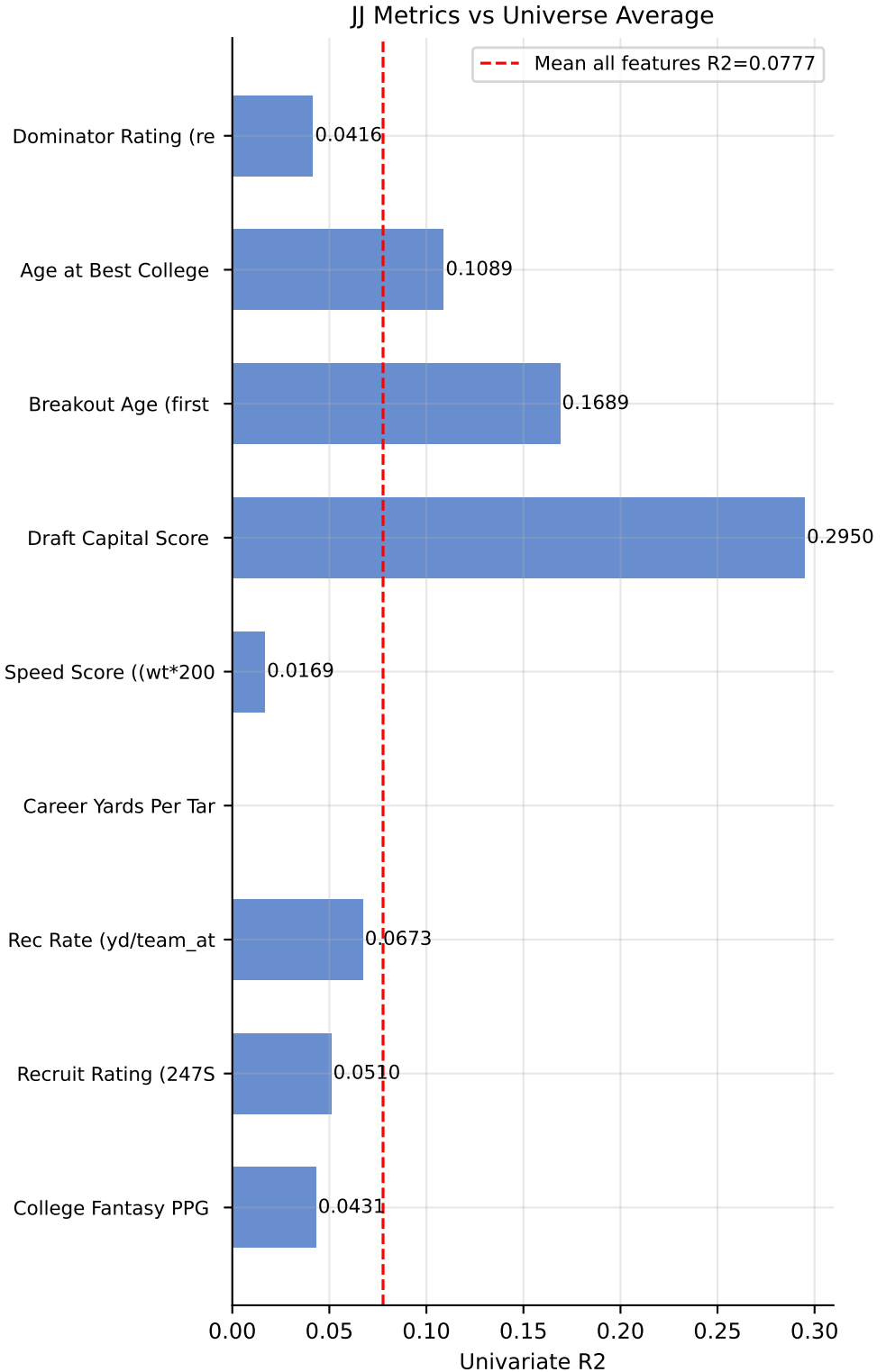




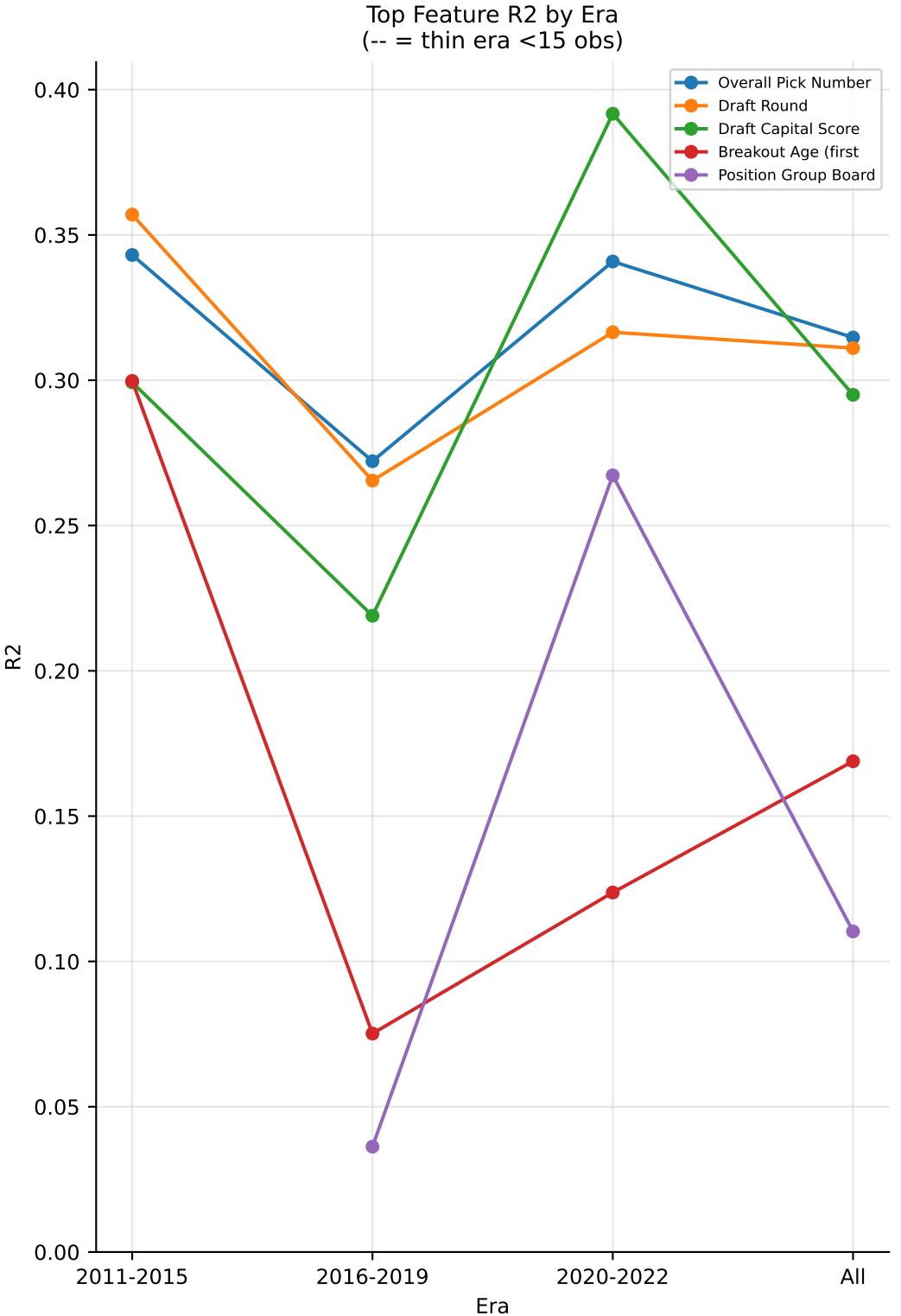
WR — JJ Zachariason Metric Benchmark vs b2s_score

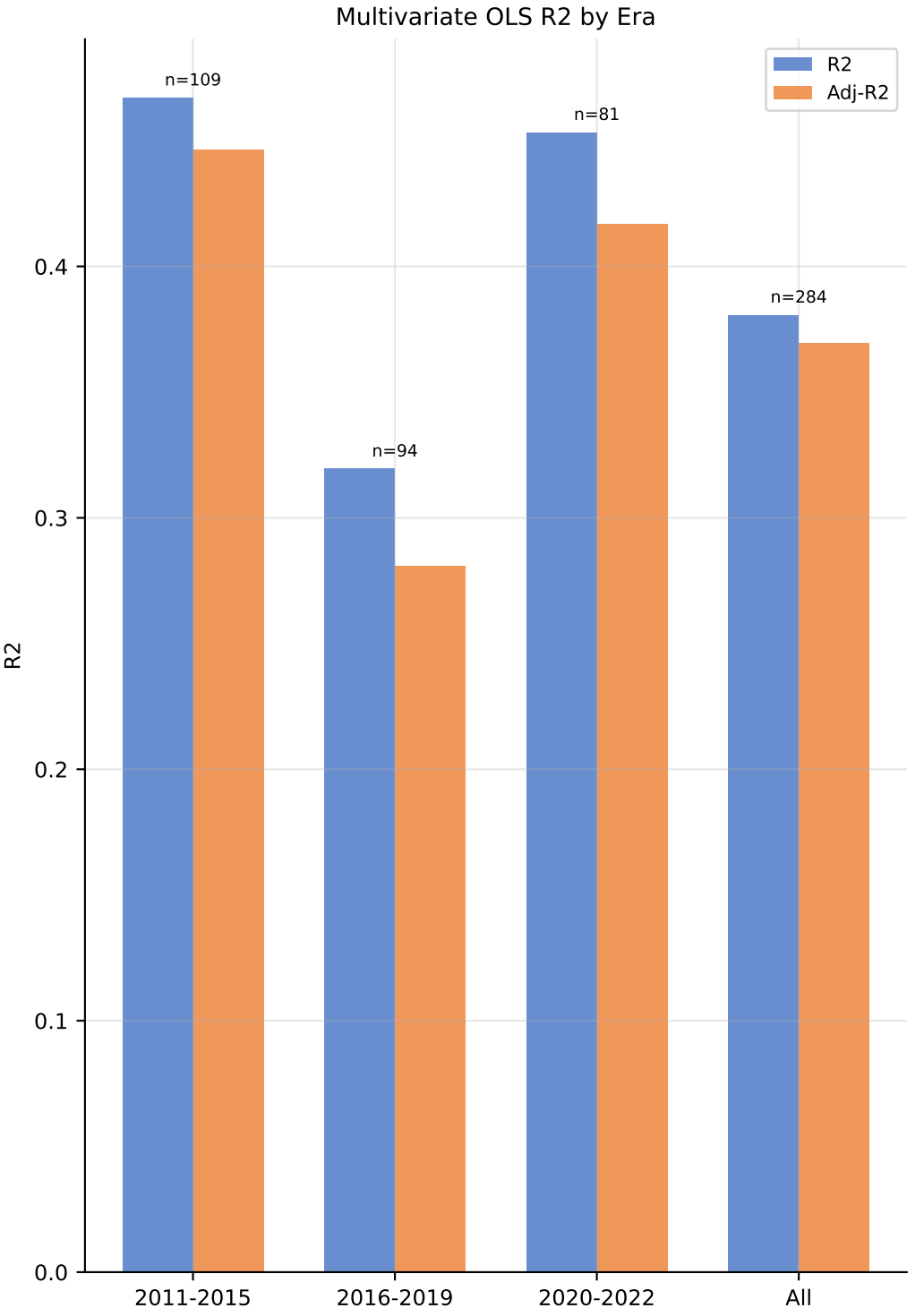
Feature	JZ Claim	Our R2	Rank	N	p-val
Dominator Ratin	>20% = strong	0.0416	32	284	0.001
Age at Best Coll	Younger at peak	0.1089	16	284	0.000
Breakout Age (fi	<20 at first 20%	0.1689	12	247	0.000
Draft Capital Sc	Strongest single	0.2950	4	284	0.000
Speed Score ((w	Most predictive	0.0169	44	218	0.056
Career Yards Pe	YPRR proxy; yar	--	71	0	--
Rec Rate (yd/tea	Rec yds / team	0.0673	18	284	0.000
Recruit Rating (2	Elite recruits sig	0.0510	25	18	0.368
College Fantasy	PPR fantasy PPG	0.0431	30	284	0.000

Green=R2>=0.05 | Yellow=R2>=0.01 | Red=R2<0.01 (weak signal in our data)



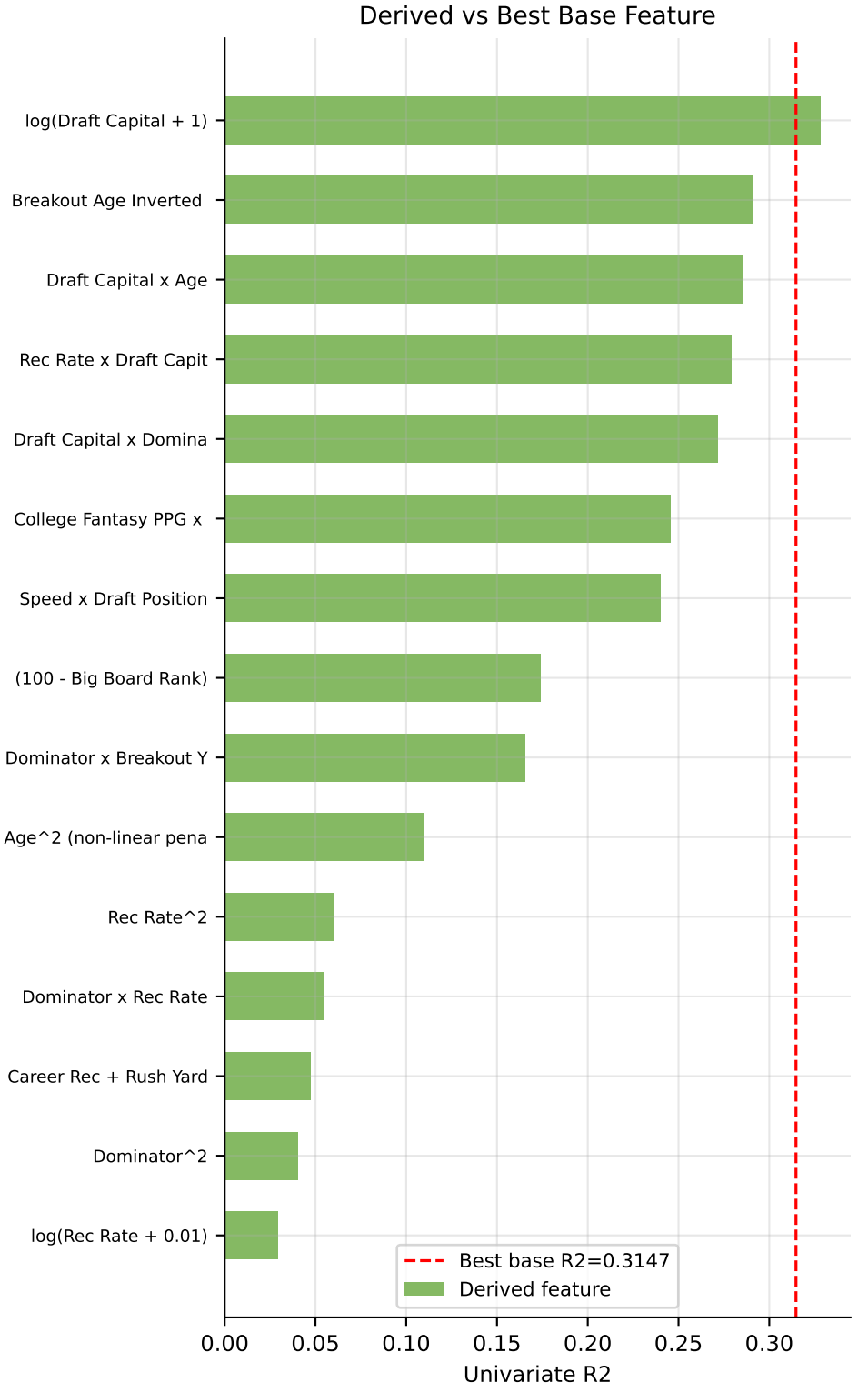
WR — Era Sensitivity Analysis

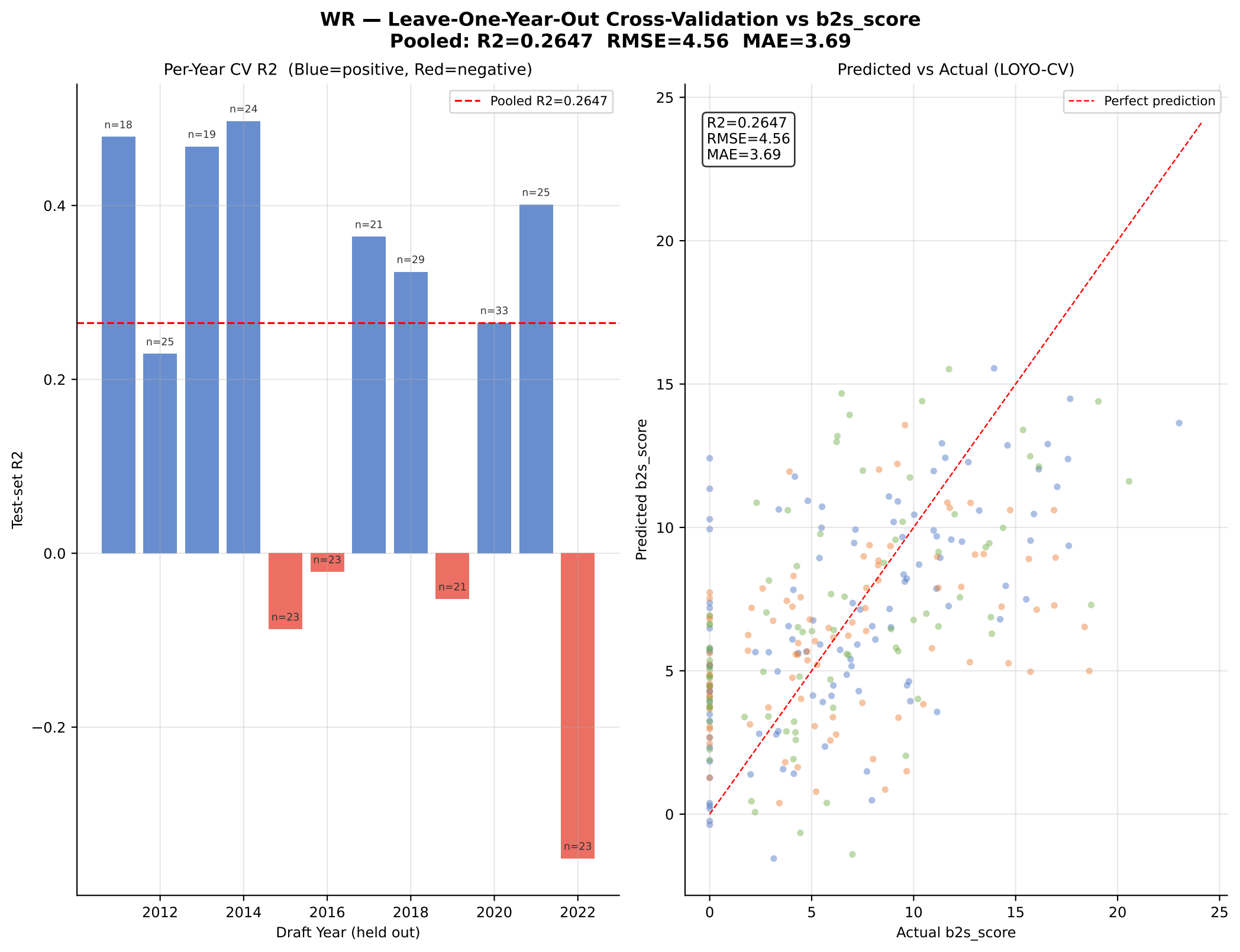




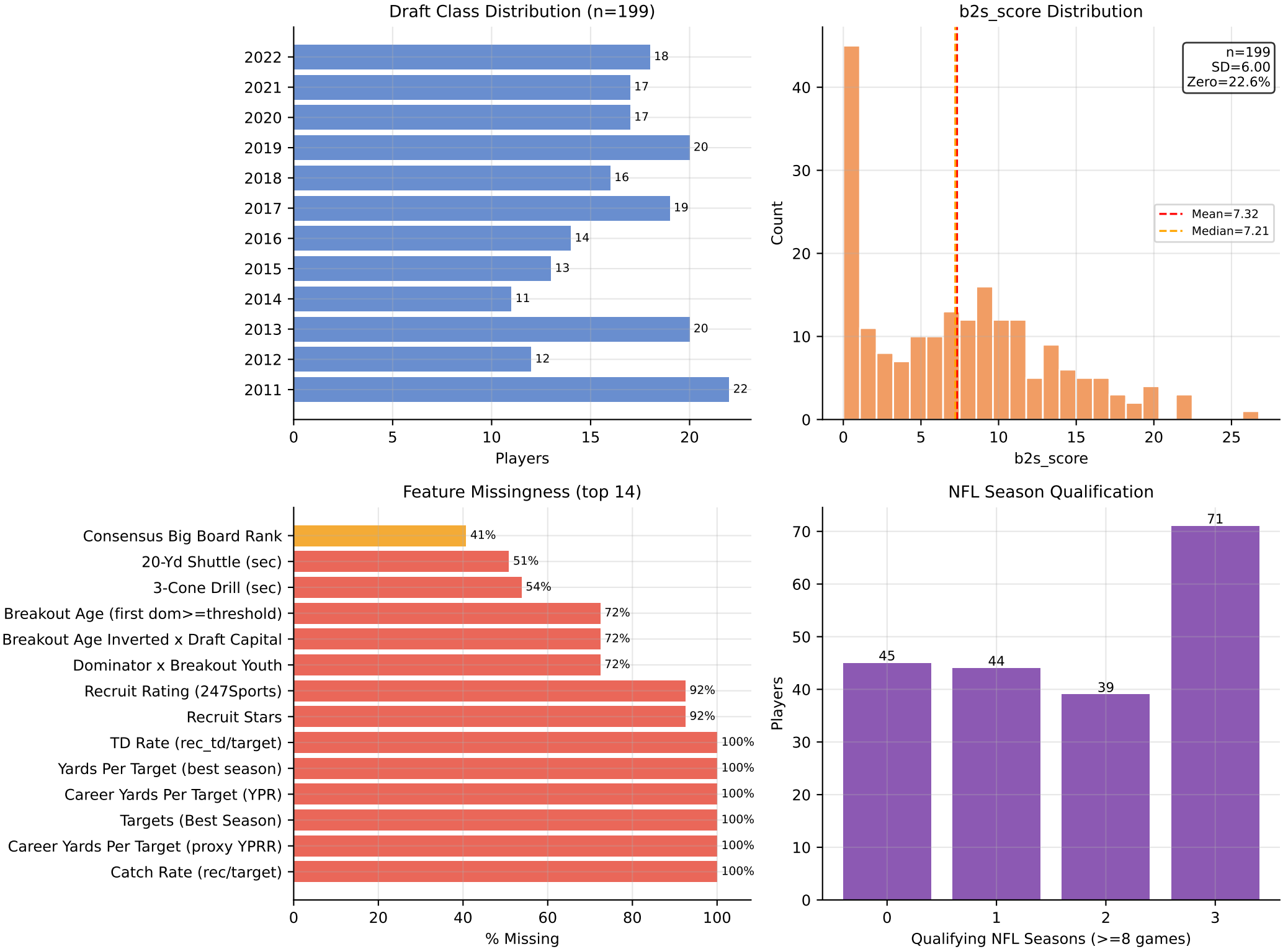
WR — Feature Engineering Results vs b2s_score

Derived Feature	R2	Best Base R2	Delta	N
log(Draft Capital + 1	0.3283	0.3147	+0.0136	284
akout Age Inverted x I	0.2908	0.3147	-0.0239	247
Draft Capital x Age	0.2857	0.3147	-0.0290	284
Rec Rate x Draft Capit	0.2792	0.3147	-0.0355	284
Draft Capital x Domina	0.2715	0.3147	-0.0432	284
lege Fantasy PPG x Dra	0.2456	0.3147	-0.0691	284
ed x Draft Position Co	0.2402	0.3147	-0.0745	284
00 - Big Board Rank) /	0.1739	0.3147	-0.1408	172
ominator x Breakout Y	0.1656	0.3147	-0.1491	247
ge^2 (non-linear pena	0.1095	0.3147	-0.2052	284
Rec Rate^2	0.0604	0.3147	-0.2543	284
Dominator x Rec Rate	0.0549	0.3147	-0.2598	284
Career Rec + Rush Yar	0.0471	0.3147	-0.2676	284
Dominator^2	0.0405	0.3147	-0.2742	284
log(Rec Rate + 0.01)	0.0294	0.3147	-0.2853	284



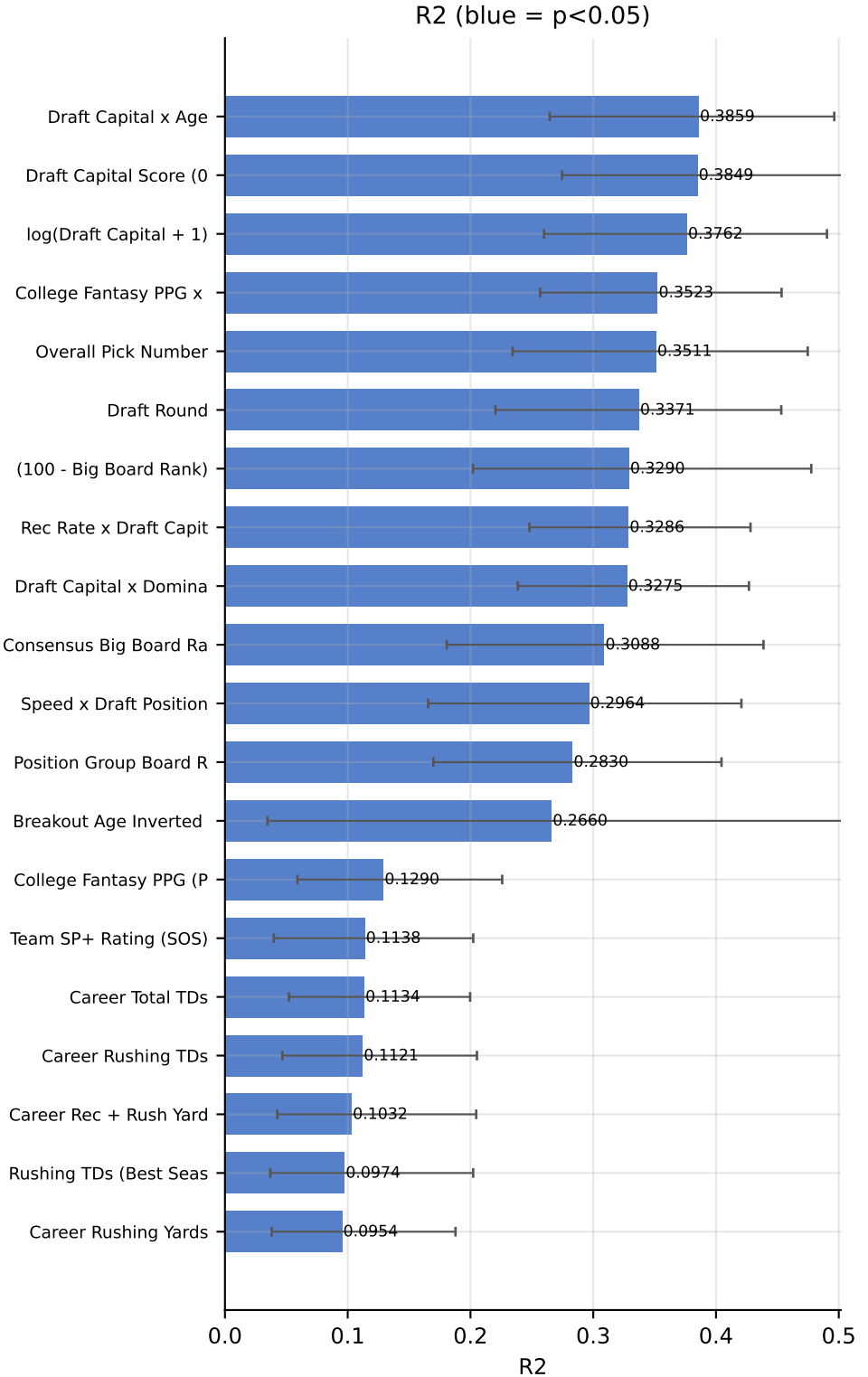

WR — Leave-One-Year-Out Cross-Validation vs b2s_score
Pooled: R2=0.2647 RMSE=4.56 MAE=3.69
Per-Year CV R2 (Blue=positive, Red=negative)
Pooled R2=0.2647
n=18
n=25
n=19
n=24
n=23
n=23
n=21
n=29
n=21
n=33
n=25
n=23
Test-set R2
Draft Year (held out)
Predicted vs Actual (LOYO-CV)
Perfect prediction
R2=0.2647
RMSE=4.56
MAE=3.69
Predicted b2s_score
Actual b2s_score

RB — Data Overview



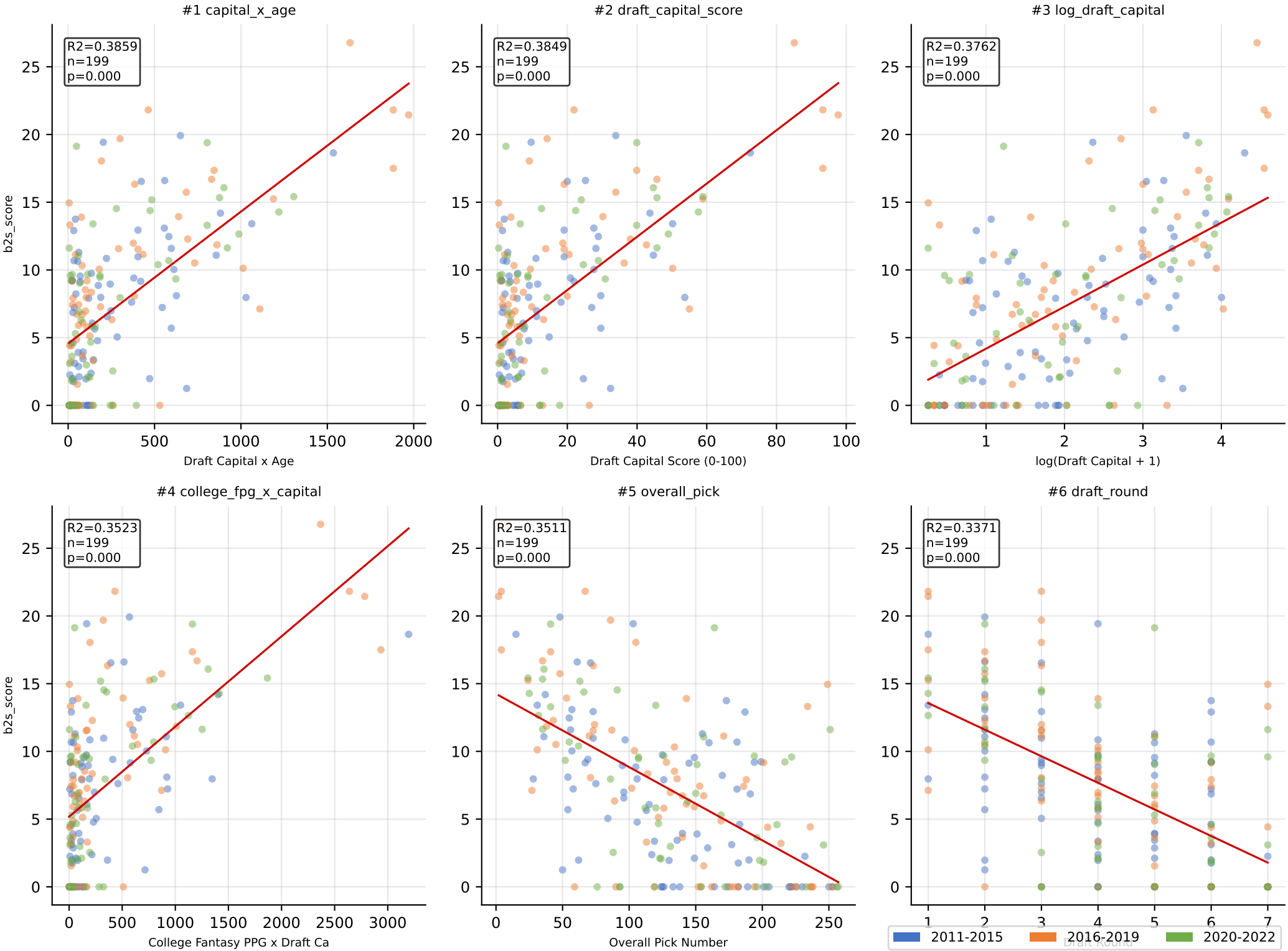
RB — Univariate R2 Ranking vs b2s_score

Rank	Feature	N	R2	95% CI	r	p-val
1	Draft Capital x Ag	199	0.3859	[0.265, 0.496]	+0.621	0.000
2	Draft Capital Score (0	199	0.3849	[0.275, 0.508]	+0.620	0.000
3	log(Draft Capital +	199	0.3762	[0.260, 0.490]	+0.613	0.000
4	College Fantasy PPG x I	199	0.3523	[0.257, 0.453]	+0.594	0.000
5	Overall Pick Numb	199	0.3511	[0.234, 0.475]	-0.593	0.000
6	Draft Round	199	0.3371	[0.220, 0.453]	-0.581	0.000
7	(100 - Big Board Rank	118	0.3290	[0.202, 0.478]	+0.574	0.000
8	Rec Rate x Draft Ca	199	0.3286	[0.248, 0.428]	+0.573	0.000
9	Draft Capital x Domi	199	0.3275	[0.239, 0.427]	+0.572	0.000
10	Consensus Big Board	118	0.3088	[0.181, 0.439]	-0.556	0.000
11	Speed x Draft Position (	199	0.2964	[0.166, 0.421]	+0.544	0.000
12	Position Group Board	118	0.2830	[0.170, 0.404]	-0.532	0.000
13	Breakout Age Inverted	55	0.2660	[0.035, 0.507]	+0.516	0.000
14	College Fantasy PPG (PI	199	0.1290	[0.059, 0.226]	+0.359	0.000
15	Team SP+ Rating (S	199	0.1138	[0.040, 0.202]	+0.337	0.000
16	Career Total TDs	199	0.1134	[0.052, 0.200]	+0.337	0.000
17	Career Rushing TI	199	0.1121	[0.047, 0.205]	+0.335	0.000
18	Career Rec + Rush Y	199	0.1032	[0.043, 0.205]	+0.321	0.000
19	Rushing TDs (Best Se	199	0.0974	[0.037, 0.202]	+0.312	0.000
20	Career Rushing Ya	199	0.0954	[0.038, 0.188]	+0.309	0.000

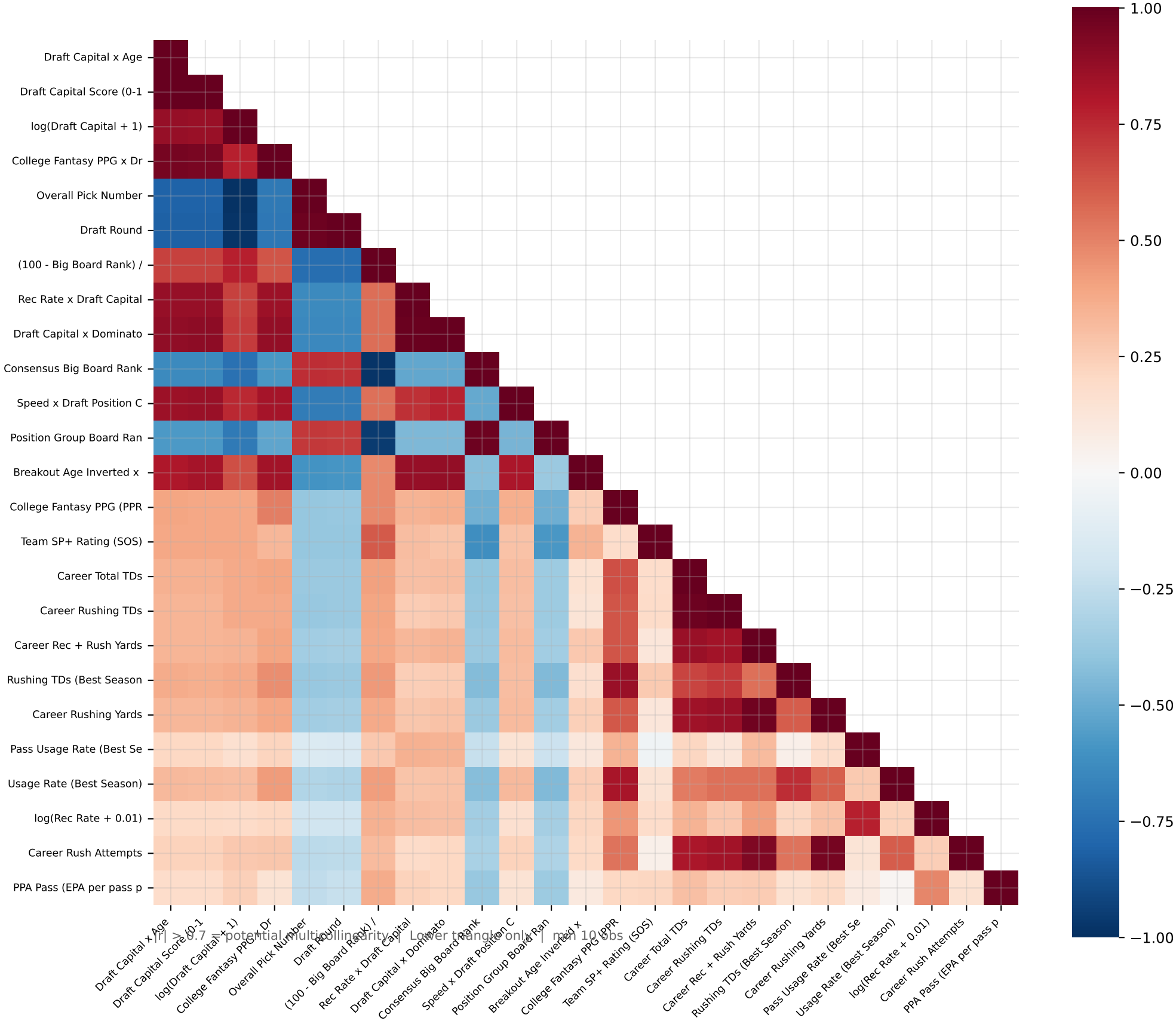


Green = R2>=0.05 & p<0.05 | Yellow = R2>=0.02 | 95% CI from 500-rep bootstrap

RB — Top Feature Scatter Plots vs b2s_score



RB — Feature Correlation Heatmap (Pearson r)

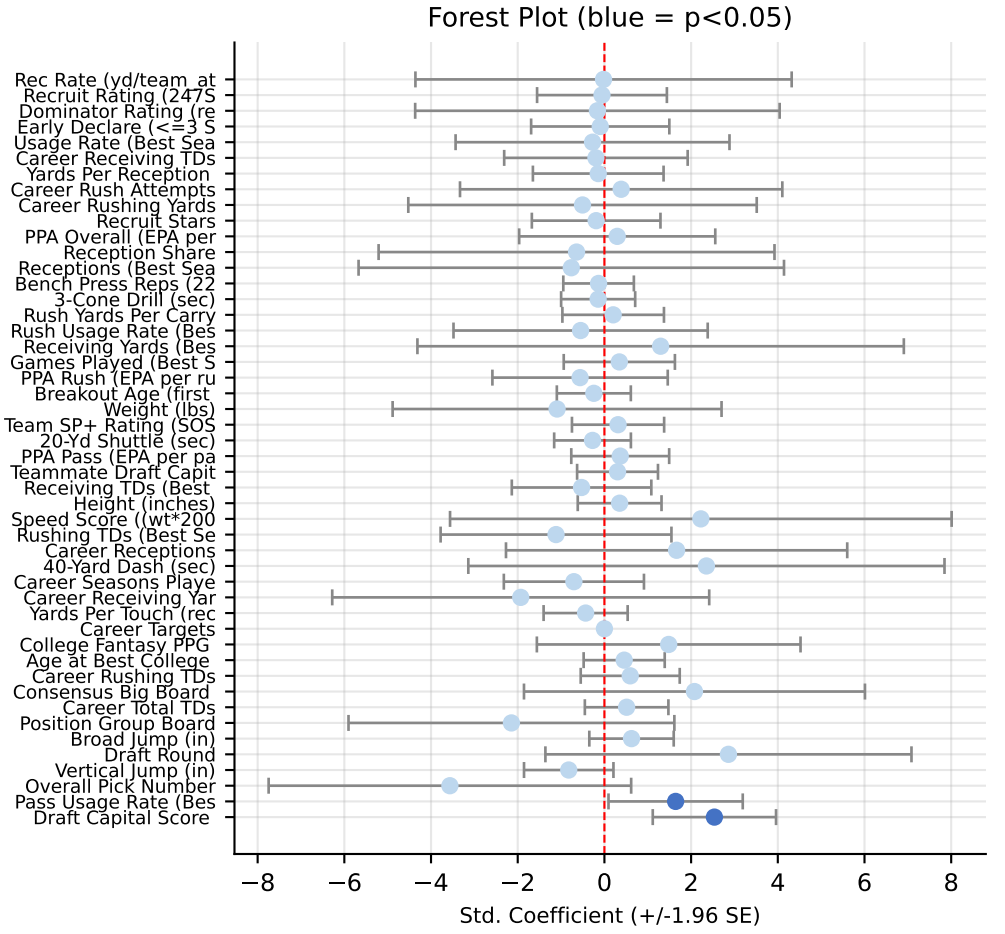


RB — OLS Regression vs b2s_score

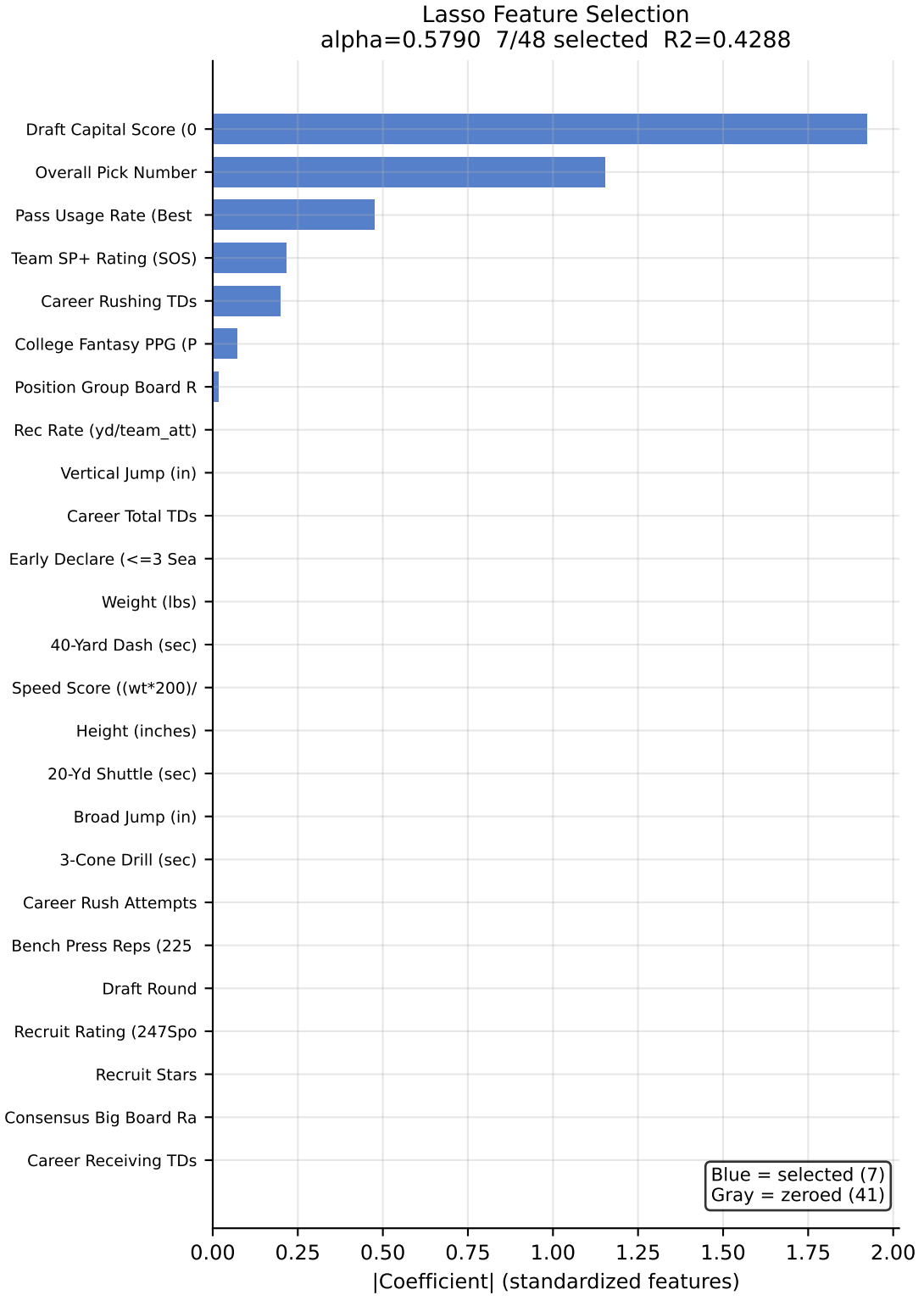
R2=0.5263 Adj-R2=0.3830 F=3.67 (p=0.0000) n=199 [Features standardized for comparability]

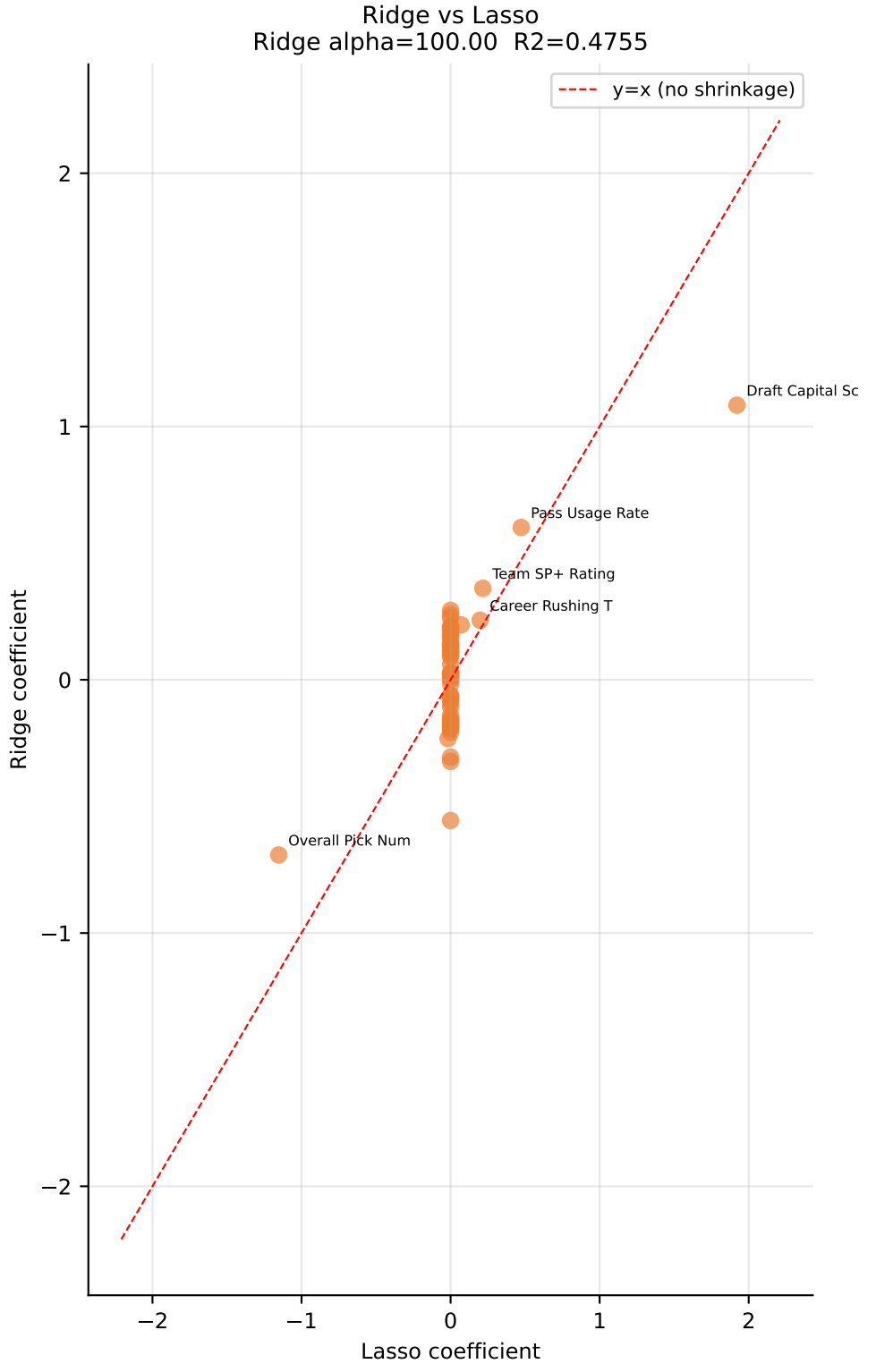
Feature	Coef	SE	t	p-val	VIF
aft Capital Score (0	+2.5373	0.7255	+3.50	0.001	4.7
s Usage Rate (Best	+1.6428	0.7900	+2.08	0.039	5.6
Overall Pick Numbe	-3.5647	2.1335	-1.67	0.097	40.8
Vertical Jump (in)	-0.8231	0.5263	-1.56	0.120	2.5
Draft Round	+2.8612	2.1544	+1.33	0.186	41.6
Broad Jump (in)	+0.6252	0.4967	+1.26	0.210	2.2
sition Group Board	-2.1450	1.9177	-1.12	0.265	33.0
Career Total TDs	+0.5123	0.4918	+1.04	0.299	inf
nsensus Big Board	+2.0776	2.0066	+1.04	0.302	36.1
Career Rushing TD	+0.5954	0.5826	+1.02	0.308	inf
e at Best College Se	+0.4561	0.4771	+0.96	0.341	2.0
llege Fantasy PPG (F	+1.4830	1.5517	+0.96	0.341	21.6
Career Targets	+0.0000	0.0000	+0.88	0.382	--
rds Per Touch (rec+r	-0.4333	0.4947	-0.88	0.382	2.2
Career Receiving Ya	-1.9309	2.2186	-0.87	0.385	44.1
Career Seasons Play	-0.7040	0.8247	-0.85	0.395	6.1
40-Yard Dash (sec	+2.3530	2.8024	+0.84	0.402	70.4
Career Receptions	+1.6655	2.0082	+0.83	0.408	36.1
shing TDs (Best Sea	-1.1163	1.3584	-0.82	0.412	16.5
ed Score ((wt*200)/	+2.2246	2.9520	+0.75	0.452	78.1
Height (inches)	+0.3512	0.4930	+0.71	0.477	2.2
ceiving TDs (Best Se	-0.5272	0.8211	-0.64	0.522	6.0
eammate Draft Cap	+0.3029	0.4759	+0.64	0.525	2.0
A Pass (EPA per pas	+0.3647	0.5761	+0.63	0.528	3.0
20-Yd Shuttle (sec	-0.2741	0.4515	-0.61	0.545	1.8
eam SP+ Rating (SO	+0.3145	0.5424	+0.58	0.563	2.6
Weight (lbs)	-1.0918	1.9357	-0.56	0.574	33.6
akout Age (first dor	-0.2441	0.4354	-0.56	0.576	1.7
A Rush (EPA per rus	-0.5616	1.0319	-0.54	0.587	9.5
mes Played (Best Se	+0.3439	0.6543	+0.53	0.600	3.8
ceiving Yards (Best	+1.2968	2.8626	+0.45	0.651	73.5
sh Usage Rate (Best	-0.5493	1.4960	-0.37	0.714	20.1
sh Yards Per Carry (	+0.2021	0.5978	+0.34	0.736	3.2
3-Cone Drill (sec)	-0.1445	0.4358	-0.33	0.741	1.7
nch Press Reps (225	-0.1345	0.4153	-0.32	0.747	1.6
eceptions (Best Sea	-0.7642	2.5037	-0.31	0.761	56.2
Reception Share	-0.6428	2.3298	-0.28	0.783	48.6
A Overall (EPA per p	+0.2953	1.1541	+0.26	0.798	11.9
Recruit Stars	-0.1905	0.7569	-0.25	0.802	5.1
Career Rushing Yar	-0.5053	2.0506	-0.25	0.806	37.7
Career Rush Attemp	+0.3863	1.8969	+0.20	0.839	32.2
rds Per Reception (R	-0.1425	0.7690	-0.19	0.853	5.3
Career Receiving TI	-0.1956	1.0799	-0.18	0.856	inf
sage Rate (Best Sea	-0.2739	1.6119	-0.17	0.865	23.3
ly Declare (<=3 Se	-0.0965	0.8134	-0.12	0.906	5.9
hinator Rating (rec_	-0.1616	2.1450	-0.08	0.940	41.2
cruit Rating (247Sp	-0.0561	0.7639	-0.07	0.942	5.2
c Rate (yd/team_att	-0.0199	2.2143	-0.01	0.993	44.0

Green=p<0.05 Yellow=p<0.10 Red VIF=>10



RB — Regularized Regression vs b2s_score

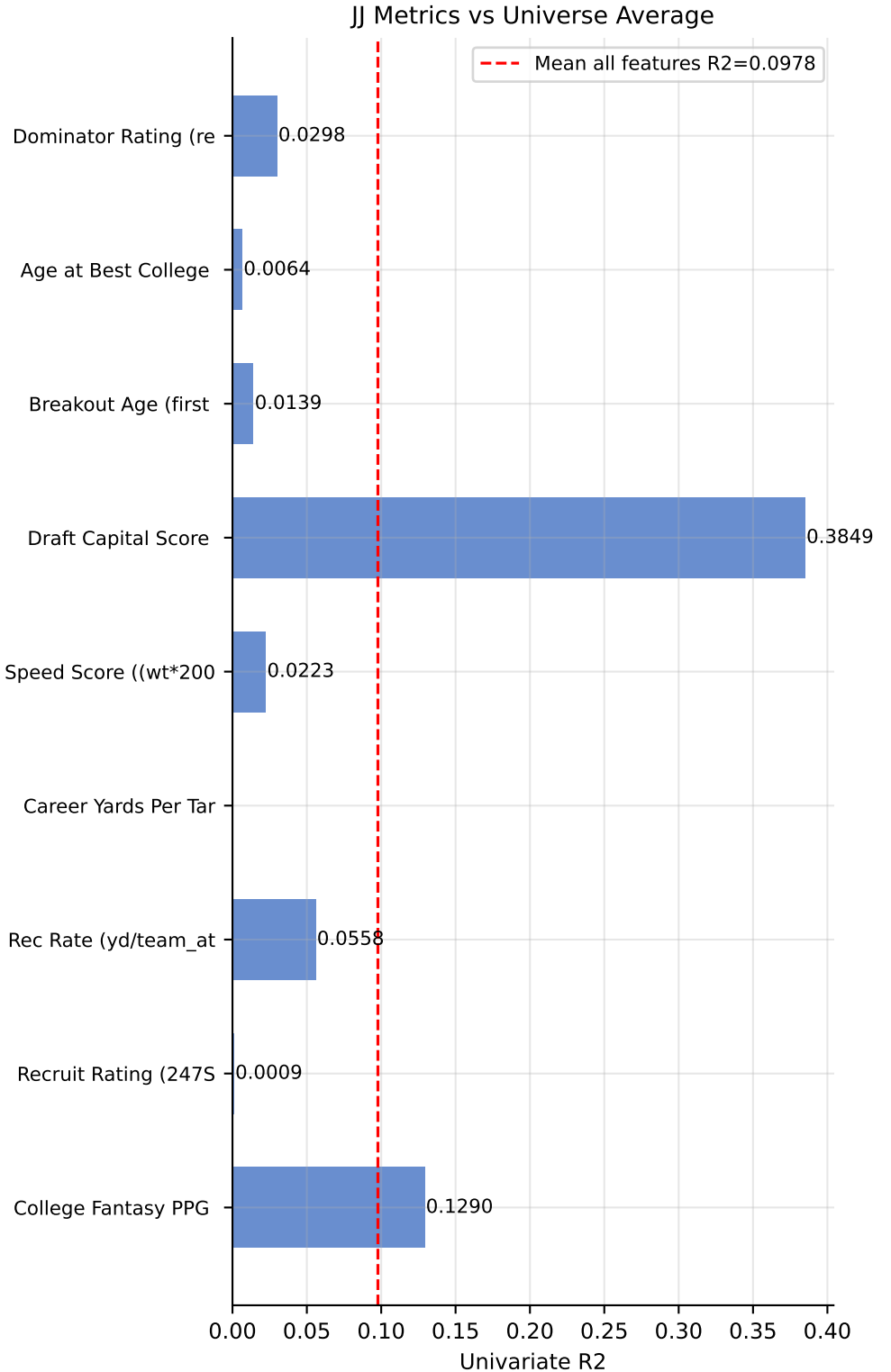




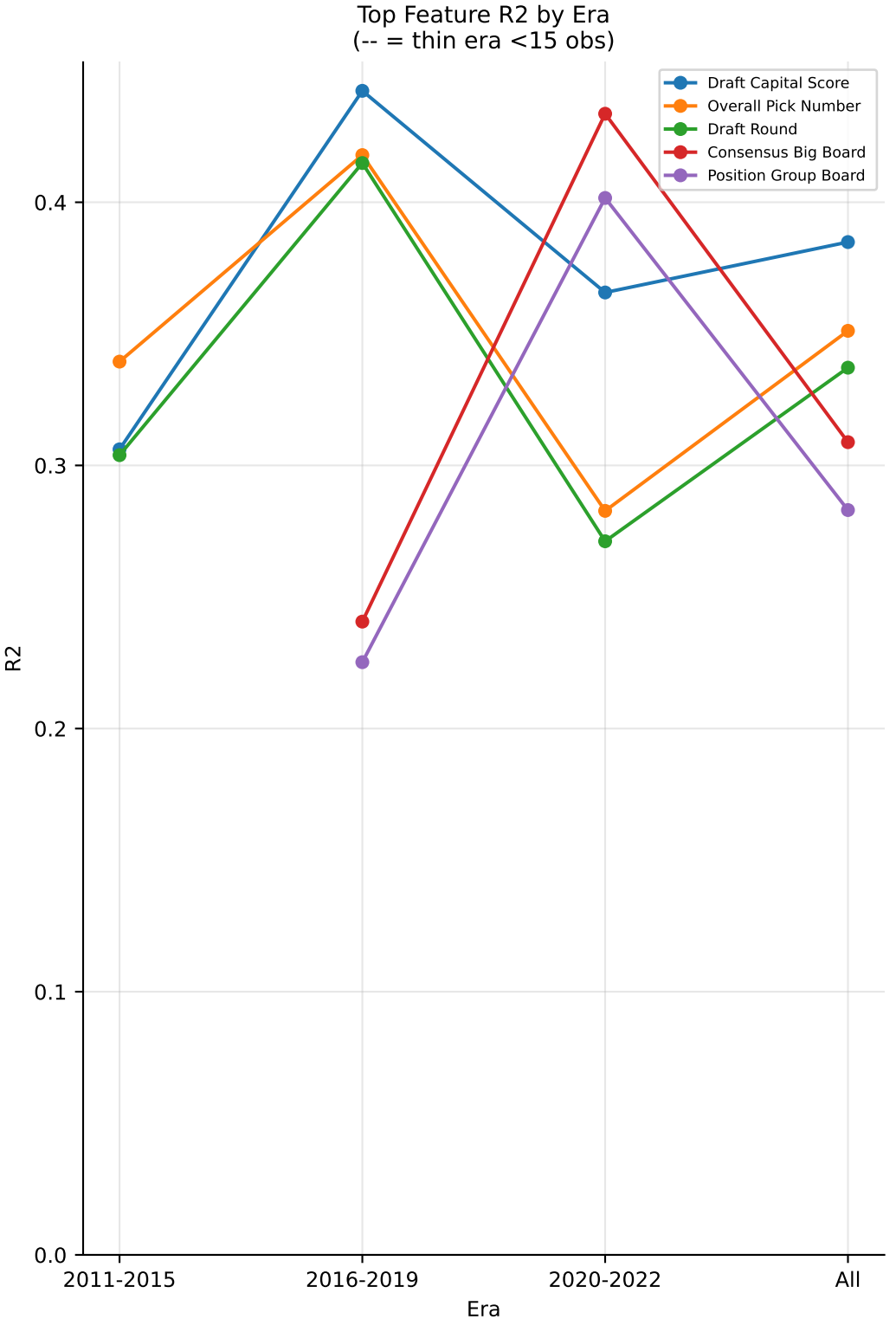
RB — JJ Zachariason Metric Benchmark vs b2s_score

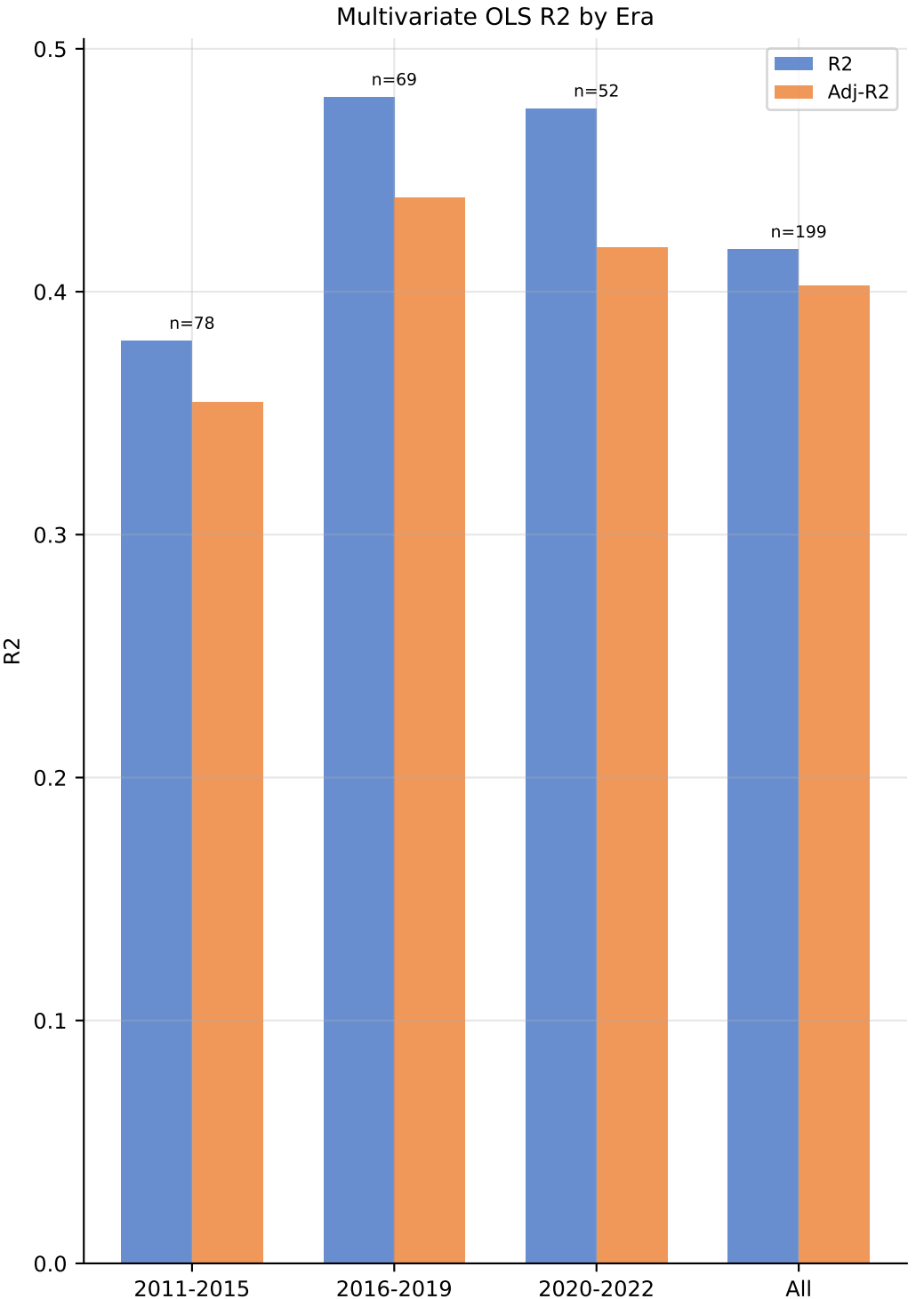
Feature	JZ Claim	Our R2	Rank	N	p-val
Dominator Ratin	>20% = strong	0.0298	36	199	0.015
Age at Best Coll	Younger at peak	0.0064	57	199	0.259
Breakout Age (fi	<20 at first 20%	0.0139	53	55	0.391
Draft Capital Sc	Strongest single	0.3849	2	199	0.000
Speed Score ((w	Most predictive	0.0223	46	157	0.062
Career Yards Pe	YPRR proxy; yar	--	71	0	--
Rec Rate (yd/tea	Rec yds / team	0.0558	26	199	0.001
Recruit Rating (2	Elite recruits sig	0.0009	63	15	0.918
College Fantasy	PPR fantasy PPG	0.1290	14	199	0.000

Green=R2>=0.05 | Yellow=R2>=0.01 | Red=R2<0.01 (weak signal in our data)



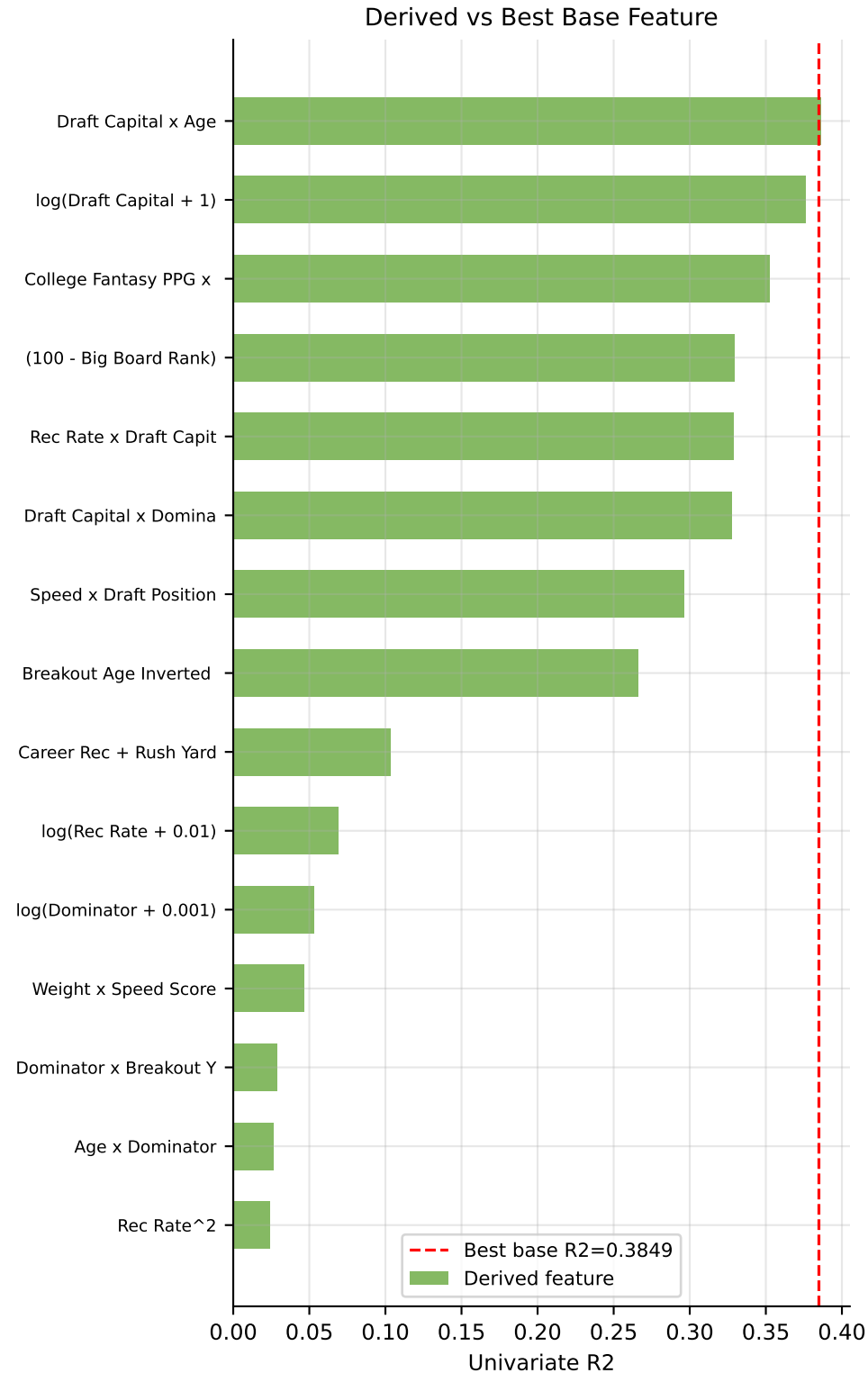
RB — Era Sensitivity Analysis

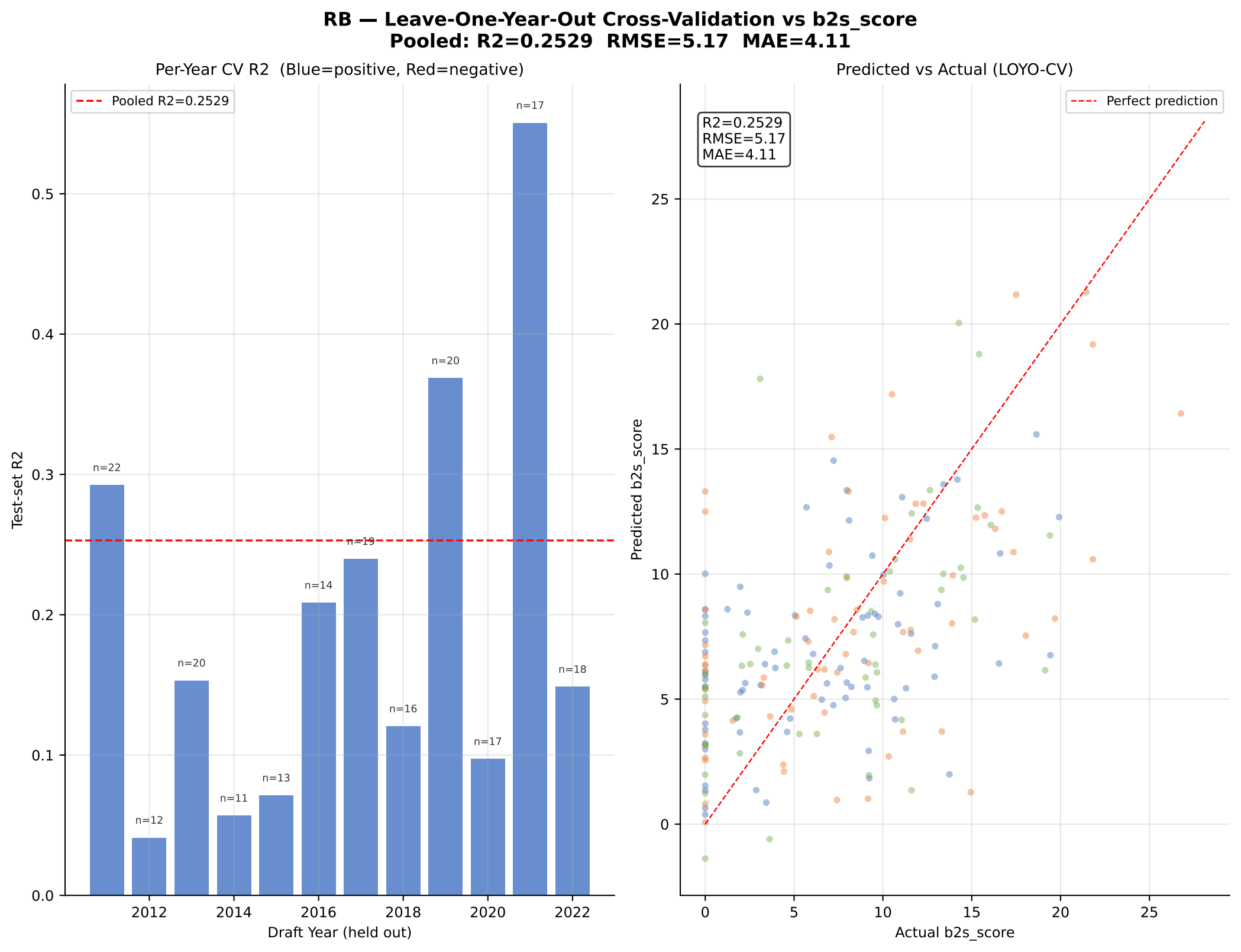




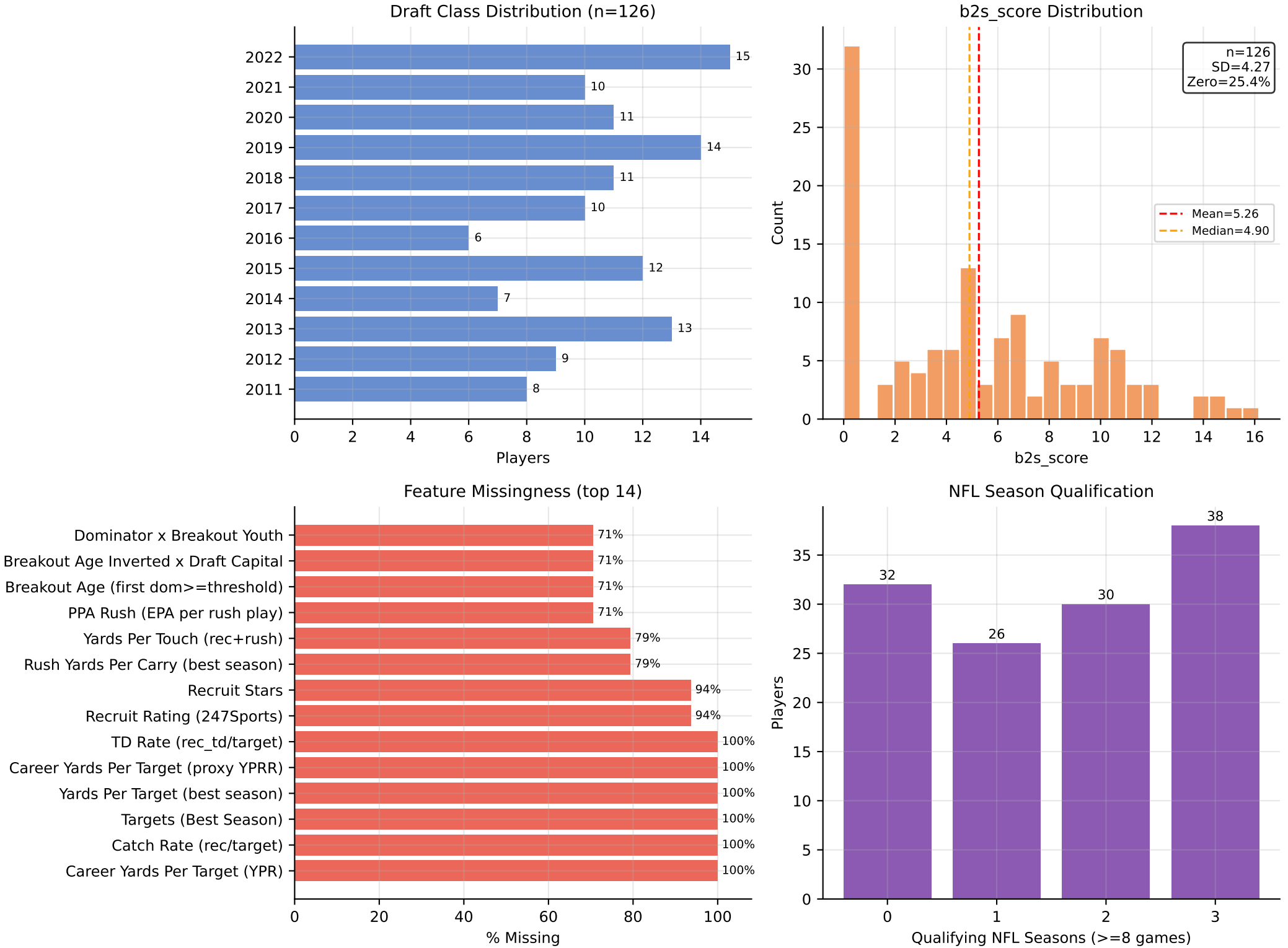
RB — Feature Engineering Results vs b2s_score

Derived Feature	R2	Best Base R2	Delta	N
Draft Capital x Age	0.3859	0.3849	+0.0010	199
log(Draft Capital + 1	0.3762	0.3849	-0.0086	199
lege Fantasy PPG x Dra	0.3523	0.3849	-0.0326	199
00 - Big Board Rank) /	0.3290	0.3849	-0.0558	118
Rec Rate x Draft Capit	0.3286	0.3849	-0.0562	199
Draft Capital x Domina	0.3275	0.3849	-0.0574	199
ed x Draft Position Co	0.2964	0.3849	-0.0885	199
akout Age Inverted x D	0.2660	0.3849	-0.1189	55
Career Rec + Rush Yar	0.1032	0.3849	-0.2817	199
log(Rec Rate + 0.01)	0.0692	0.3849	-0.3157	199
log(Dominator + 0.00	0.0527	0.3849	-0.3322	199
Weight x Speed Score	0.0466	0.3849	-0.3383	157
ominator x Breakout Y	0.0290	0.3849	-0.3559	55
Age x Dominator	0.0265	0.3849	-0.3584	199
Rec Rate^2	0.0238	0.3849	-0.3611	199



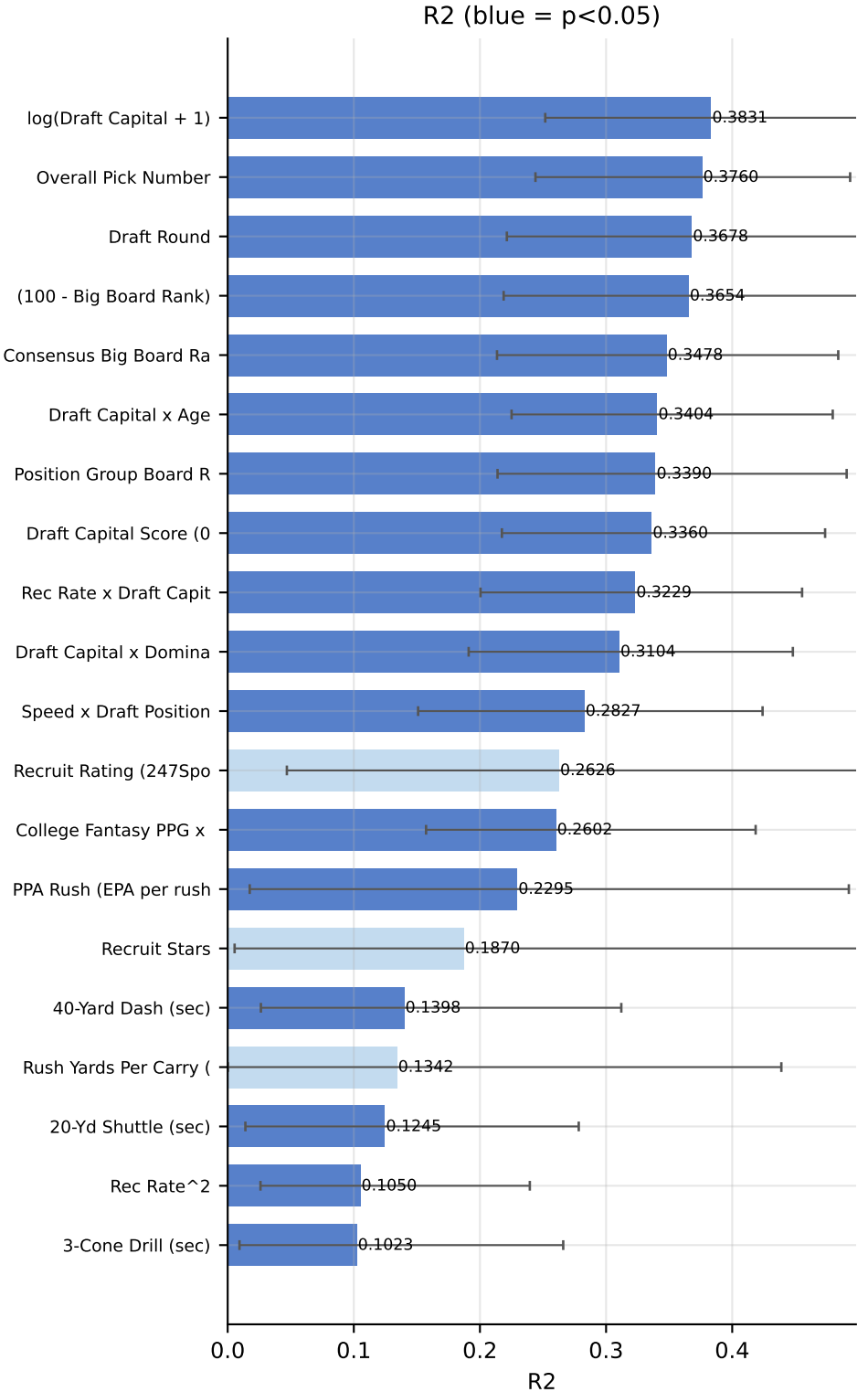

RB — Leave-One-Year-Out Cross-Validation vs b2s_score
Pooled: R2=0.2529 RMSE=5.17 MAE=4.11
Per-Year CV R2 (Blue=positive, Red=negative)
Pooled R2=0.2529
Test-set R2
Draft Year (held out)
n=22
n=12
n=20
n=11
n=13
n=14
n=19
n=16
n=20
n=17
n=17
n=18
Predicted vs Actual (LOYO-CV)
R2=0.2529
RMSE=5.17
MAE=4.11
Perfect prediction
Predicted b2s_score
Actual b2s_score

TE — Data Overview



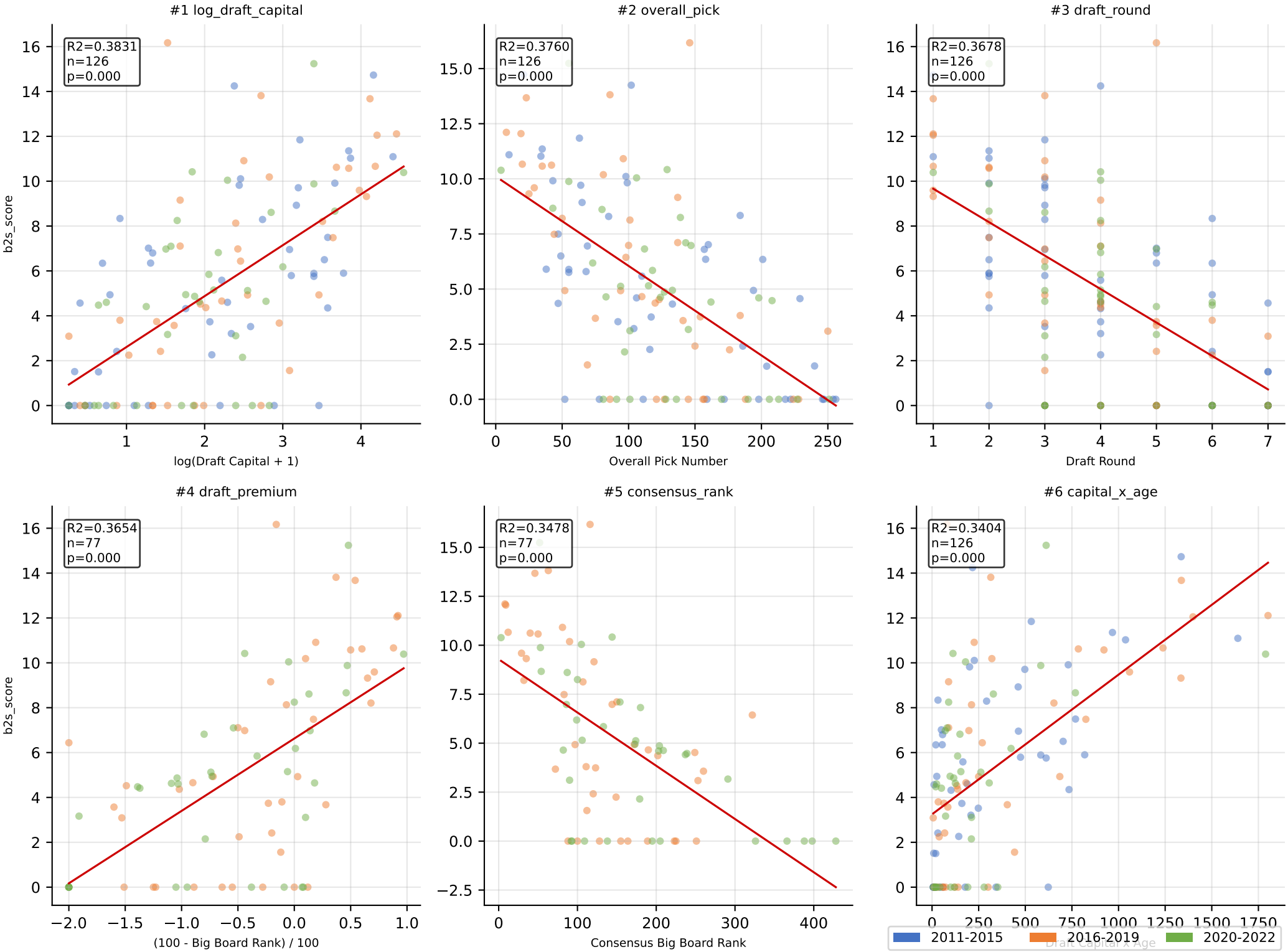
TE — Univariate R2 Ranking vs b2s_score

Rank	Feature	N	R2	95% CI	r	p-val
1	log(Draft Capital +	126	0.3831	[0.252, 0.525]	+0.619	0.000
2	Overall Pick Numb	126	0.3760	[0.244, 0.493]	-0.613	0.000
3	Draft Round	126	0.3678	[0.221, 0.527]	-0.606	0.000
4	(100 - Big Board Rank	77	0.3654	[0.219, 0.516]	+0.604	0.000
5	Consensus Big Board	77	0.3478	[0.214, 0.484]	-0.590	0.000
6	Draft Capital x Ag	126	0.3404	[0.225, 0.480]	+0.583	0.000
7	Position Group Board	77	0.3390	[0.214, 0.491]	-0.582	0.000
8	Draft Capital Score (0	126	0.3360	[0.217, 0.474]	+0.580	0.000
9	Rec Rate x Draft Ca	126	0.3229	[0.200, 0.455]	+0.568	0.000
10	Draft Capital x Domi	126	0.3104	[0.191, 0.448]	+0.557	0.000
11	Speed x Draft Position (	126	0.2827	[0.151, 0.424]	+0.532	0.000
12	Recruit Rating (247Sp	8	0.2626	[0.047, 0.659]	-0.512	0.194
13	College Fantasy PPG x I	126	0.2602	[0.157, 0.419]	+0.510	0.000
14	PPA Rush (EPA per rus	37	0.2295	[0.017, 0.492]	+0.479	0.003
15	Recruit Stars	8	0.1870	[0.005, 0.835]	-0.432	0.285
16	40-Yard Dash (sec	95	0.1398	[0.026, 0.312]	-0.374	0.000
17	Rush Yards Per Carry (b	26	0.1342	[0.000, 0.439]	+0.366	0.066
18	20-Yd Shuttle (se	76	0.1245	[0.014, 0.278]	-0.353	0.002
19	Rec Rate^2	126	0.1050	[0.026, 0.240]	+0.324	0.000
20	3-Cone Drill (sec	74	0.1023	[0.009, 0.266]	-0.320	0.005

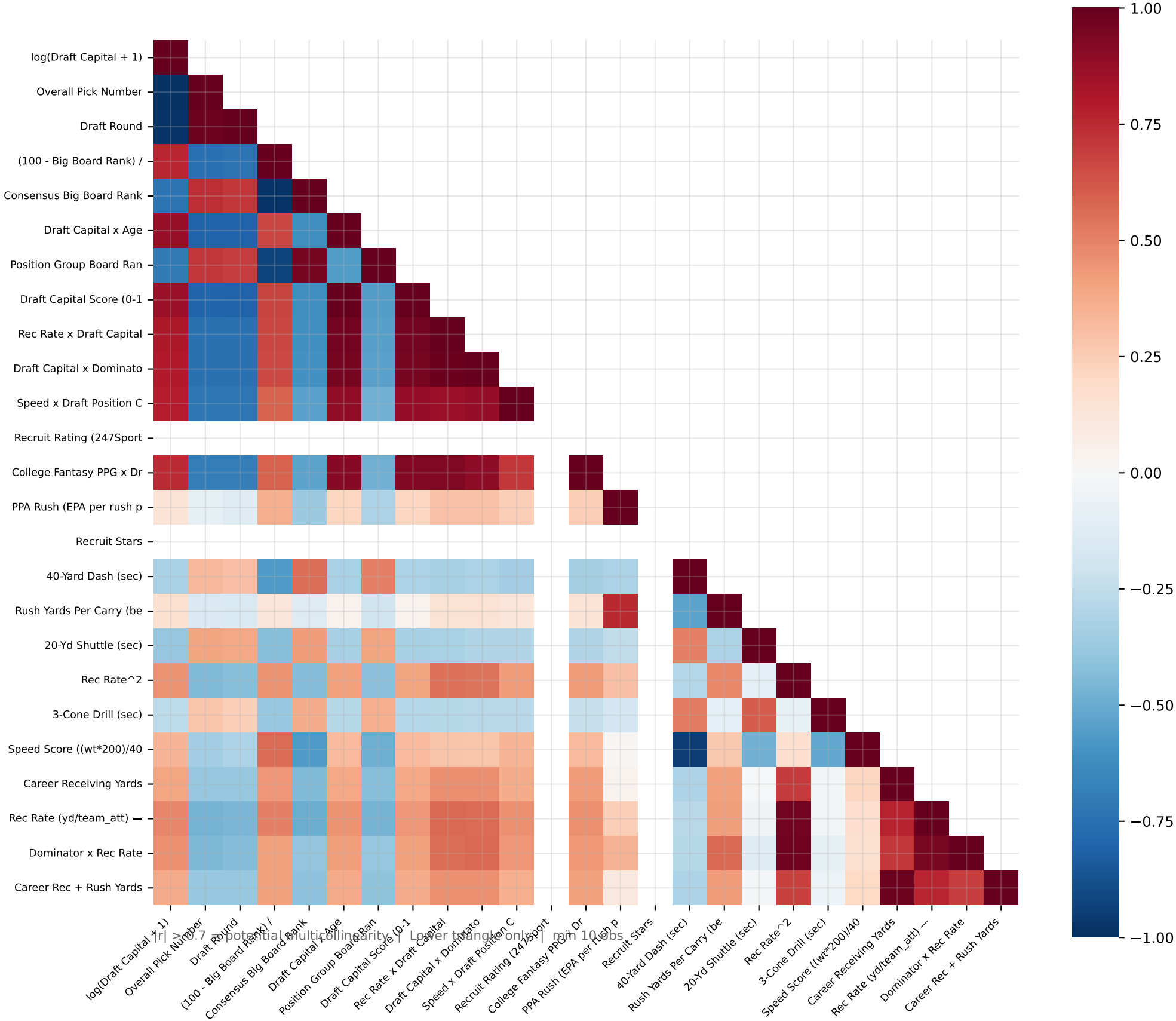


Green = R2>=0.05 & p<0.05 | Yellow = R2>=0.02 | 95% CI from 500-rep bootstrap

TE — Top Feature Scatter Plots vs b2s_score



TE — Feature Correlation Heatmap (Pearson r)



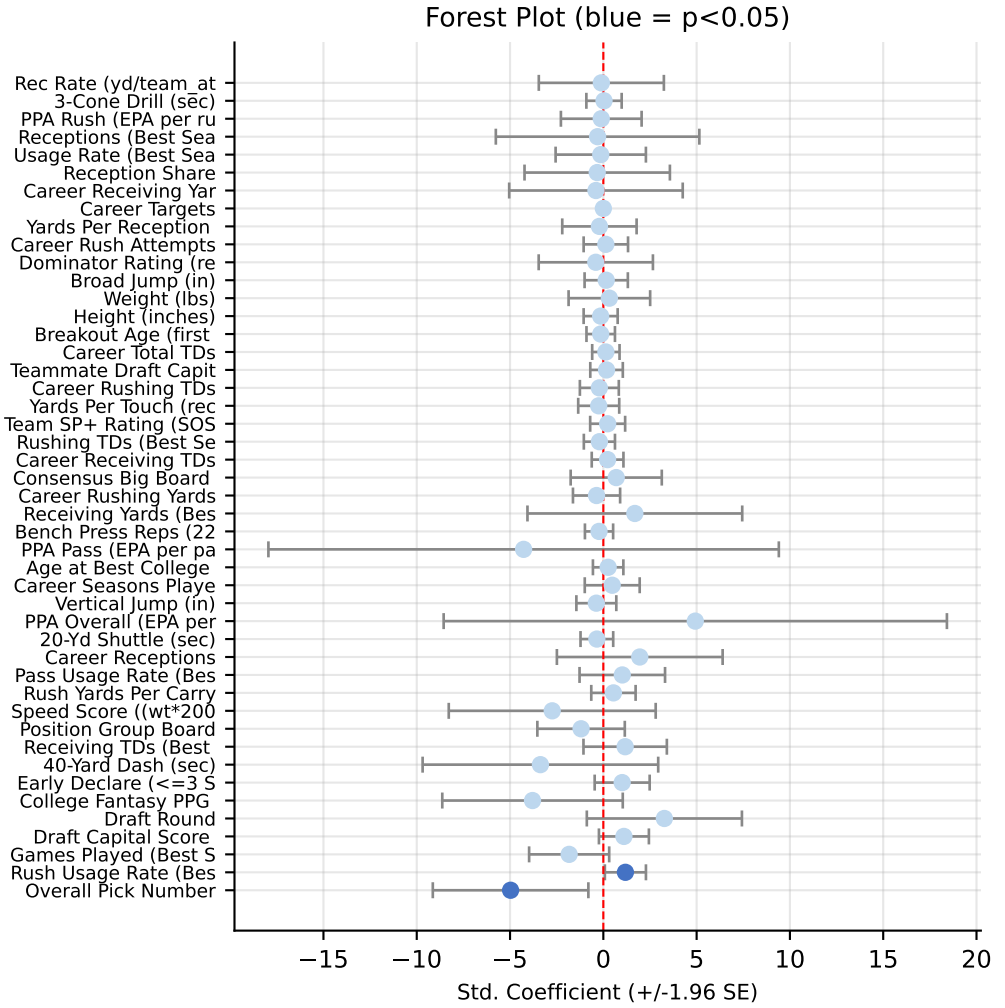
|r| ≥ 0.7 = potential multicollinearity | Lower triangle only | min 10 obs

TE — OLS Regression vs b2s_score

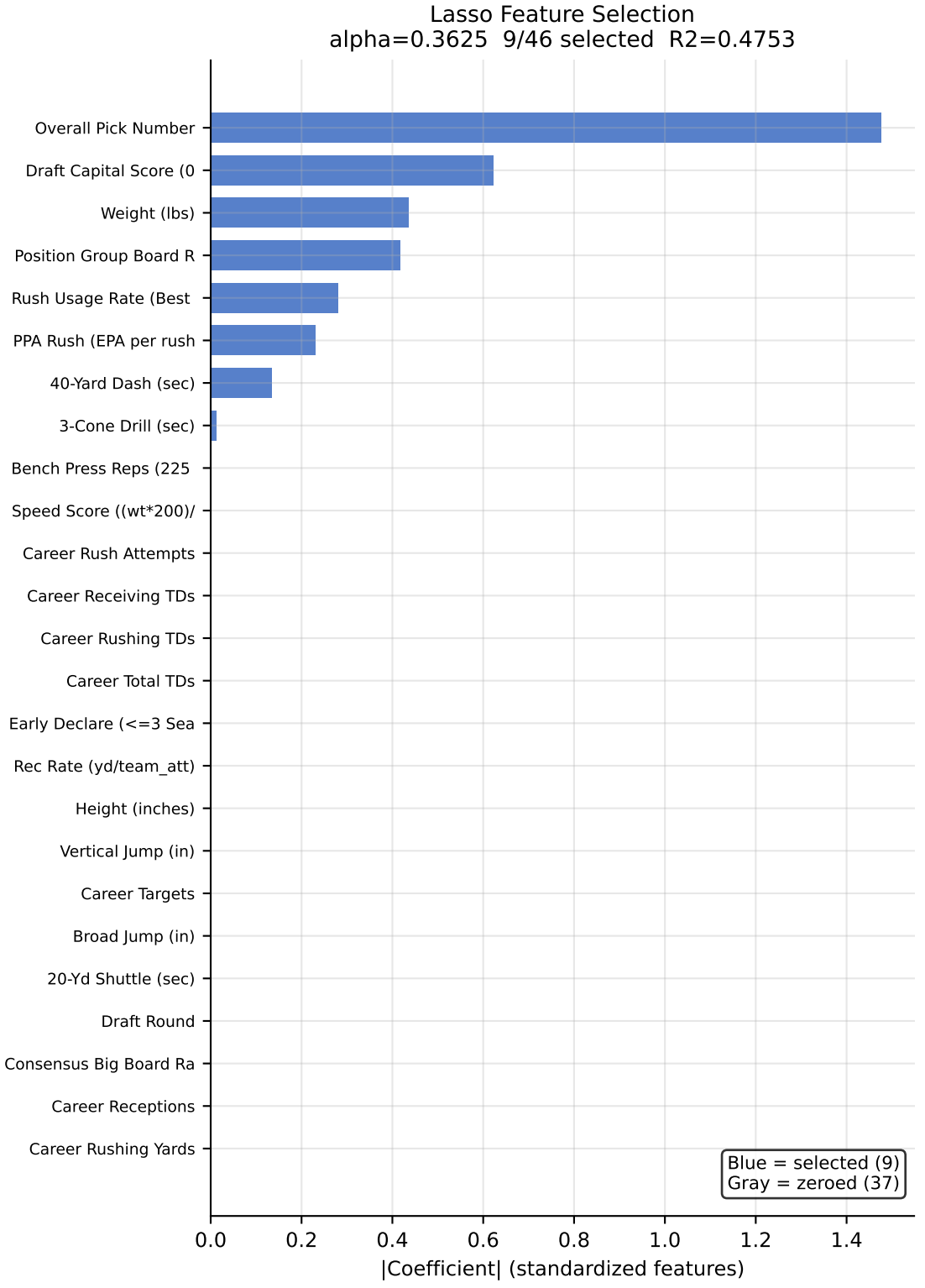
R2=0.6122 Adj-R2=0.4016 F=2.91 (p=0.0000) n=126 [Features standardized for comparability]

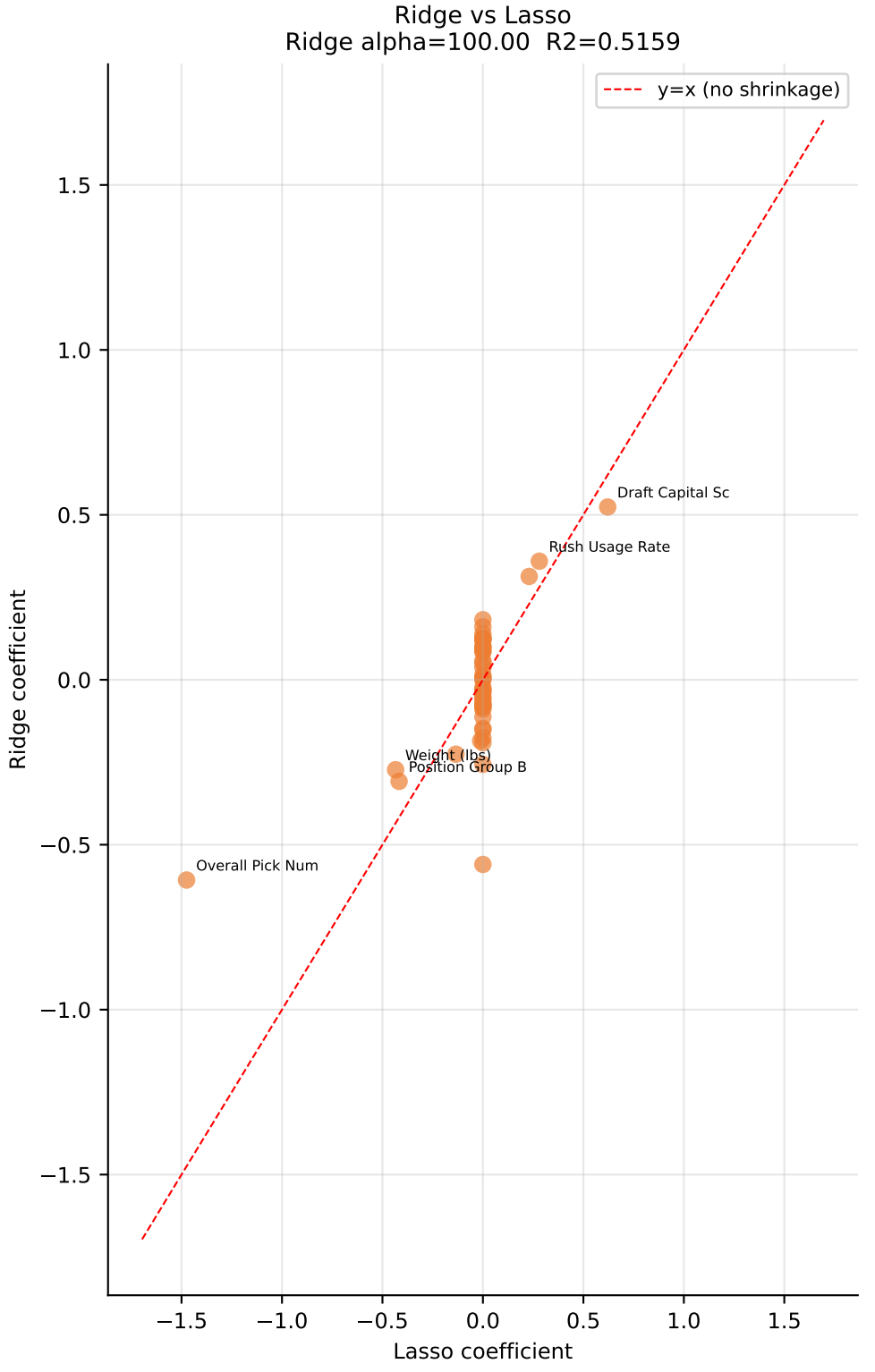
Feature	Coef	SE	t	p-val	VIF
Overall Pick Numbe	-4.9707	2.1269	-2.34	0.022	52.1
h Usage Rate (Best	+1.1827	0.5616	+2.11	0.038	3.6
nes Played (Best Se	-1.8321	1.0948	-1.67	0.098	13.8
aft Capital Score (0	+1.1032	0.6830	+1.62	0.110	5.4
Draft Round	+3.2712	2.1207	+1.54	0.127	51.8
lege Fantasy PPG (F	-3.7952	2.4677	-1.54	0.128	70.2
ly Declare (<=3 Se	+1.0097	0.7518	+1.34	0.183	6.5
40-Yard Dash (sec	-3.3721	3.2209	-1.05	0.298	119.6
ceiving TDs (Best Se	+1.1699	1.1400	+1.03	0.308	15.0
sition Group Board	-1.1930	1.1955	-1.00	0.321	16.5
ed Score ((wt*200)	-2.7390	2.8292	-0.97	0.336	92.3
sh Yards Per Carry (	+0.5417	0.6064	+0.89	0.374	4.2
s Usage Rate (Best	+1.0164	1.1689	+0.87	0.387	15.8
Career Receptions	+1.9524	2.2662	+0.86	0.391	59.2
20-Yd Shuttle (sec	-0.3468	0.4468	-0.78	0.440	2.3
A Overall (EPA per p	+4.9283	6.8805	+0.72	0.476	545.7
Vertical Jump (in)	-0.3748	0.5460	-0.69	0.494	3.4
Career Seasons Play	+0.4766	0.7510	+0.63	0.527	6.5
e at Best College Se	+0.2581	0.4161	+0.62	0.537	2.0
A Pass (EPA per pas	-4.2721	6.9785	-0.61	0.542	561.4
nch Press Reps (225	-0.2307	0.3855	-0.60	0.551	1.7
ceiving Yards (Best	+1.6888	2.9363	+0.58	0.567	99.4
Career Rushing Yar	-0.3641	0.6448	-0.56	0.574	4.8
nsensus Big Board	+0.6910	1.2457	+0.55	0.581	17.9
Career Receiving TI	+0.2280	0.4334	+0.53	0.600	inf
shing TDs (Best Sea	-0.2105	0.4257	-0.49	0.622	2.1
eam SP+ Rating (SO	+0.2317	0.4781	+0.48	0.629	2.6
rds Per Touch (rec+r	-0.2476	0.5607	-0.44	0.660	3.6
Career Rushing TD	-0.2139	0.5322	-0.40	0.689	inf
eammate Draft Cap	+0.1721	0.4449	+0.39	0.700	2.3
Career Total TDs	+0.1360	0.3741	+0.36	0.717	inf
akout Age (first dor	-0.1403	0.3895	-0.36	0.720	1.8
Height (inches)	-0.1434	0.4652	-0.31	0.759	2.5
Weight (lbs)	+0.3235	1.1151	+0.29	0.772	14.3
Broad Jump (in)	+0.1548	0.5915	+0.26	0.794	4.0
nimator Rating (rec_	-0.4033	1.5618	-0.26	0.797	28.1
Career Rush Attemp	+0.1342	0.6070	+0.22	0.826	4.2
rds Per Reception (F	-0.2090	1.0166	-0.21	0.838	11.9
Career Targets	+0.0000	0.0000	+0.17	0.862	--
Career Receiving Ya	-0.3952	2.3726	-0.17	0.868	64.9
Reception Share	-0.3281	1.9876	-0.17	0.869	45.5
age Rate (Best Sea	-0.1383	1.2345	-0.11	0.911	17.6
eceptions (Best Sea	-0.3078	2.7826	-0.11	0.912	89.3
A Rush (EPA per rus	-0.1115	1.1036	-0.10	0.920	14.0
3-Cone Drill (sec)	+0.0368	0.4828	+0.08	0.939	2.7
c Rate (yd/team_att	-0.1066	1.7101	-0.06	0.950	33.7

Green=p<0.05 Yellow=p<0.10 Red VIF=>10



TE — Regularized Regression vs b2s_score

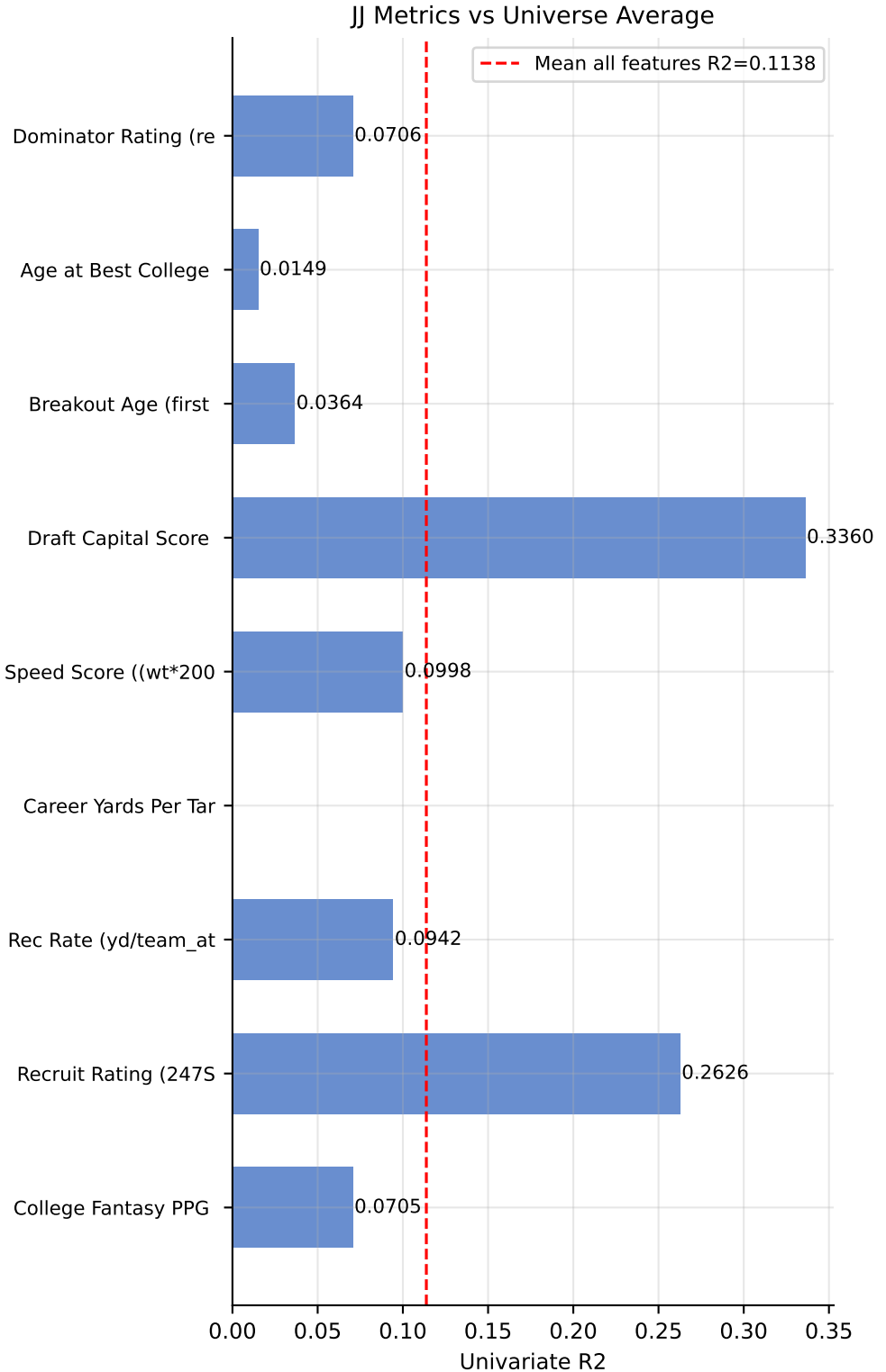




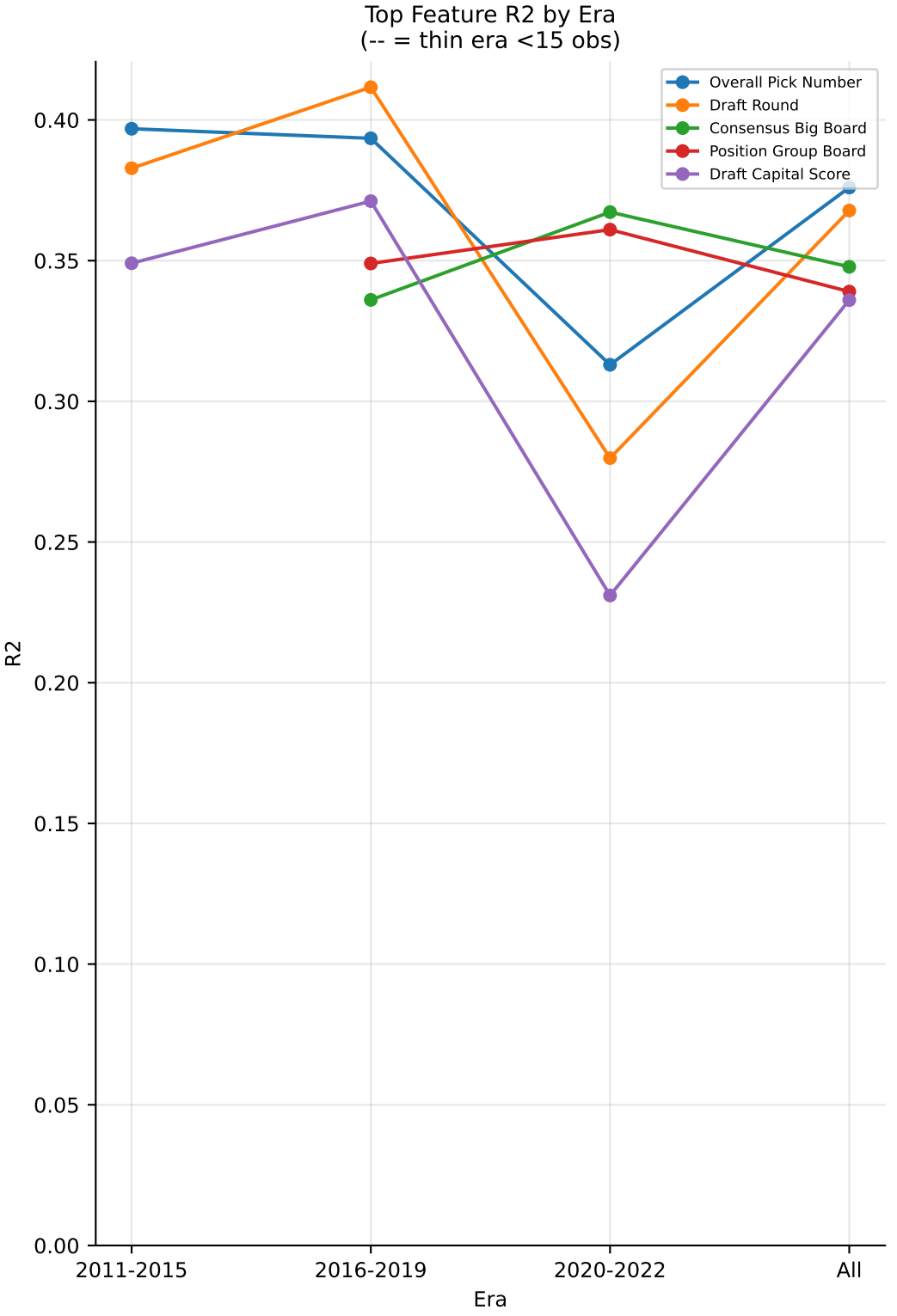
TE — JJ Zachariason Metric Benchmark vs b2s_score

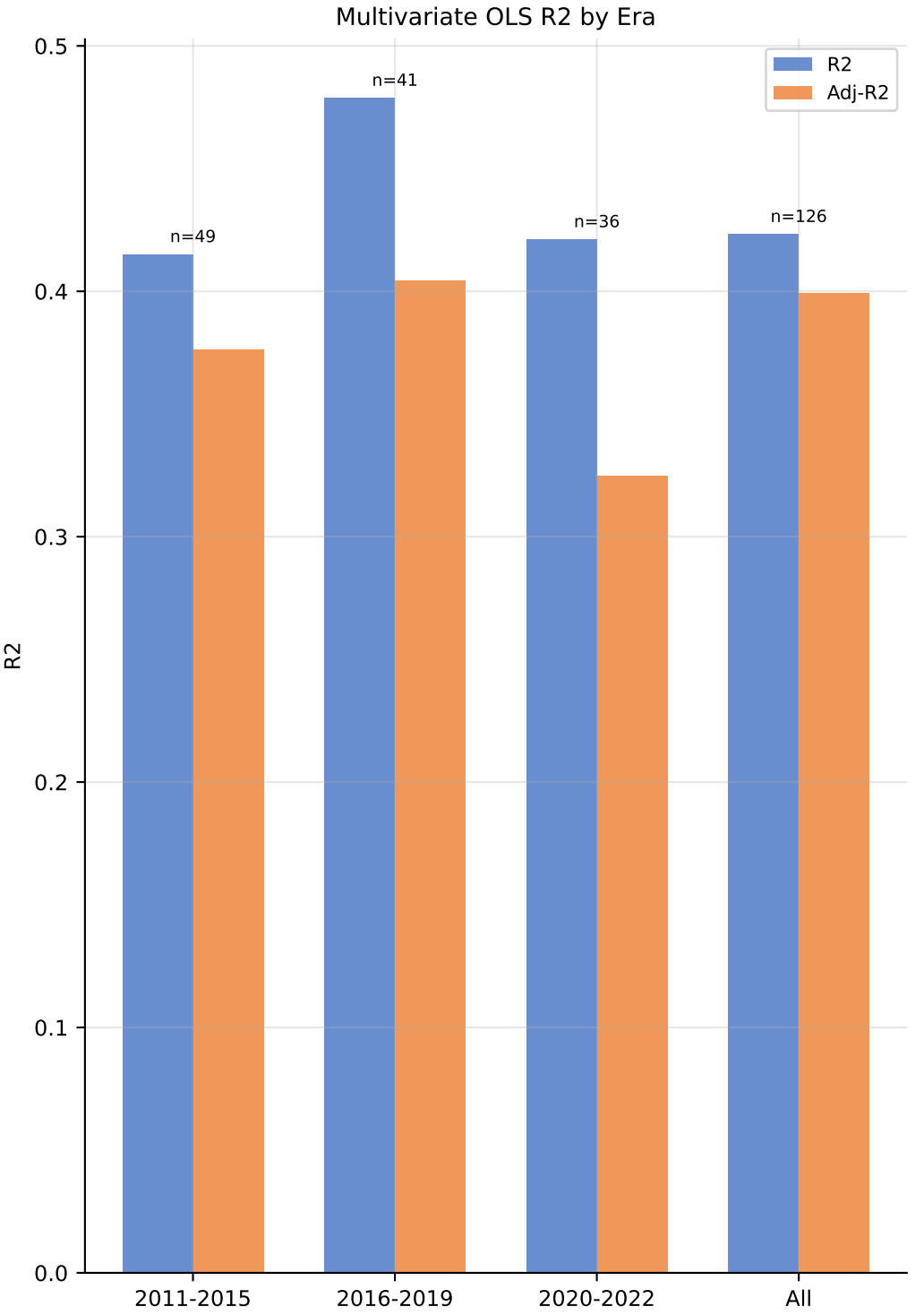
Feature	JZ Claim	Our R2	Rank	N	p-val
Dominator Ratin	>20% = strong	0.0706	31	126	0.003
Age at Best Coll	Younger at peak	0.0149	53	126	0.173
Breakout Age (fi	<20 at first 20%	0.0364	48	37	0.258
Draft Capital Sc	Strongest single	0.3360	8	126	0.000
Speed Score ((w	Most predictive	0.0998	21	95	0.002
Career Yards Pe	YPRR proxy; yar	--	71	0	--
Rec Rate (yd/tea	Rec yds / team	0.0942	23	126	0.000
Recruit Rating (2	Elite recruits sig	0.2626	12	8	0.194
College Fantasy	PPR fantasy PPG	0.0705	32	126	0.003

Green=R2>=0.05 | Yellow=R2>=0.01 | Red=R2<0.01 (weak signal in our data)



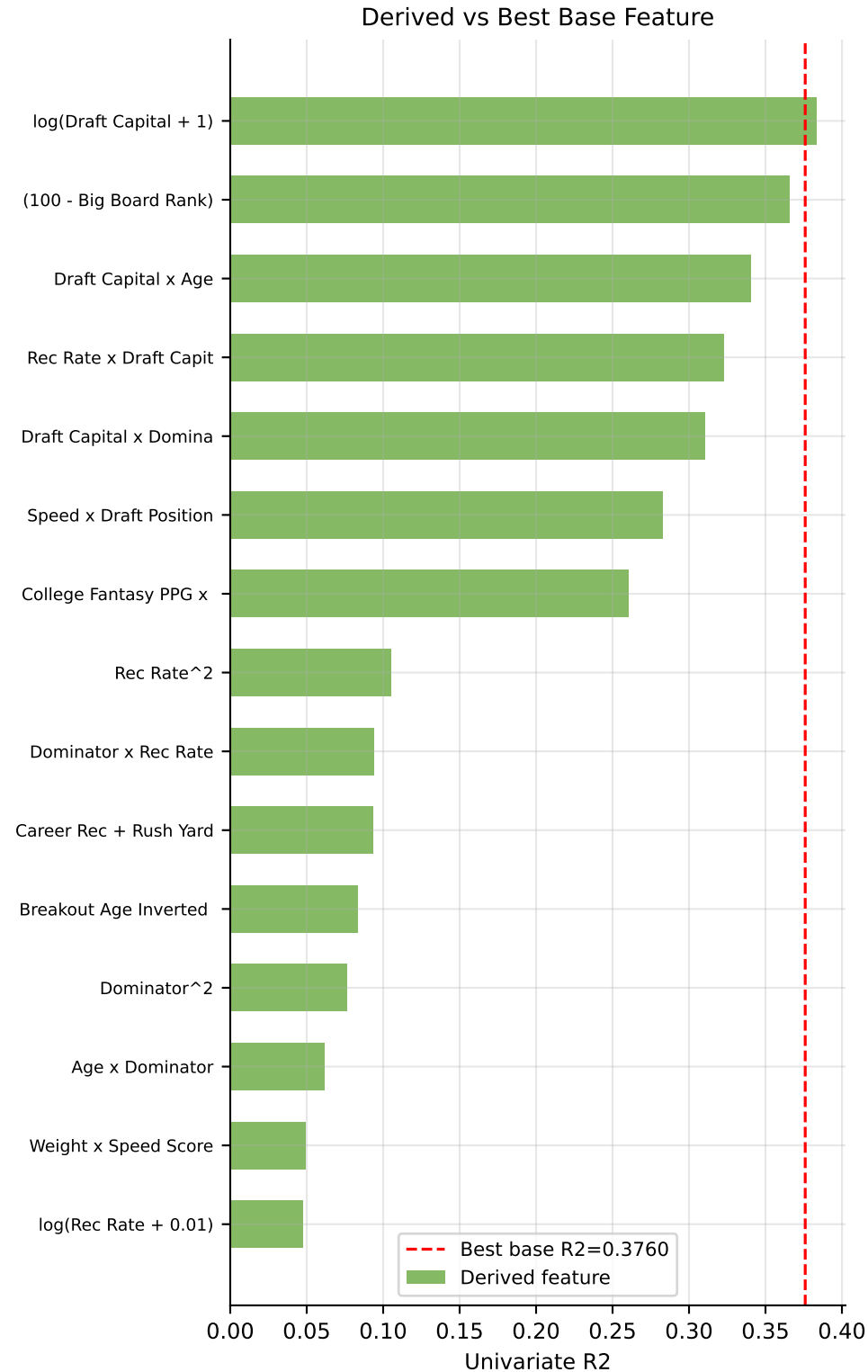
TE — Era Sensitivity Analysis

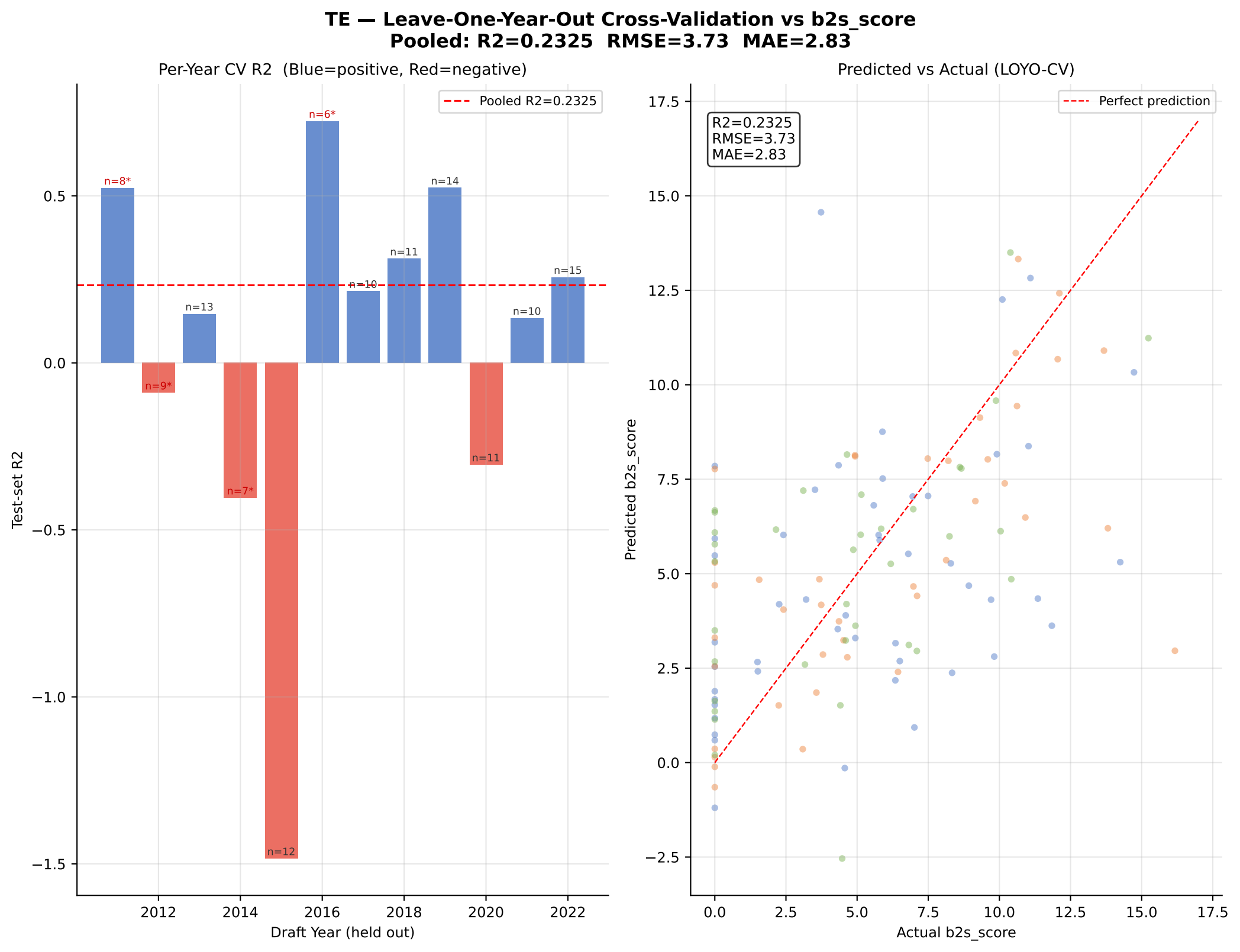




TE — Feature Engineering Results vs b2s_score

Derived Feature	R2	Best Base R2	Delta	N
log(Draft Capital + 1	0.3831	0.3760	+0.0072	126
00 - Big Board Rank) /	0.3654	0.3760	-0.0106	77
Draft Capital x Age	0.3404	0.3760	-0.0356	126
Rec Rate x Draft Capit	0.3229	0.3760	-0.0531	126
Draft Capital x Domina	0.3104	0.3760	-0.0656	126
ed x Draft Position Co	0.2827	0.3760	-0.0933	126
lege Fantasy PPG x Dra	0.2602	0.3760	-0.1158	126
Rec Rate^2	0.1050	0.3760	-0.2709	126
Dominator x Rec Rate	0.0939	0.3760	-0.2821	126
Career Rec + Rush Yar	0.0932	0.3760	-0.2828	126
akout Age Inverted x D	0.0830	0.3760	-0.2929	37
Dominator^2	0.0762	0.3760	-0.2998	126
Age x Dominator	0.0613	0.3760	-0.3147	126
Weight x Speed Score	0.0490	0.3760	-0.3270	95
log(Rec Rate + 0.01)	0.0475	0.3760	-0.3285	126




TE — Leave-One-Year-Out Cross-Validation vs b2s_score
Pooled: R2=0.2325 RMSE=3.73 MAE=2.83
Per-Year CV R2 (Blue=positive, Red=negative)
Pooled R2=0.2325
n=8*
n=9*
n=13
n=7*
n=12
n=6*
n=10
n=11
n=14
n=11
n=10
n=15
Test-set R2
Draft Year (held out)
Predicted vs Actual (LOYO-CV)
Perfect prediction
R2=0.2325
RMSE=3.73
MAE=2.83
Predicted b2s_score
Actual b2s_score

Missing Data Inventory & Improvement Roadmap

```
DATA GAPS — WHAT IS MISSING

1. recruit_rating (~80% missing)
   Blocked: CFBD monthly API quota exhausted 2026-02-22.
   Fix: resume in March 2026 with populate_db.py
        --recruit-years 2011 2012 2013 2014 2015 2016 2017

2. consensus_rank (~41% missing)
   Structural: no free source exists pre-2016.
   Fix: accept gap, or scrape archived ESPN boards manually.

3. speed_score / forty_time (~21-25% missing)
   Cause: combine non-attenders (NOT random — lower-upside
   players more likely to skip). Consider indicator variable.

4. best_ppa_pass / best_usage (early CFBD gap)
   Cause: CFBD did not track PPA/usage in 2011-2013 seasons.

5. College routes run (YPRR) — entirely unavailable free
   career_rec_per_target is a weak proxy only.
   Fix: PFF+ (~$200/yr) for true YPRR back to 2014.

6. NFL-level metrics — permanently unavailable free:
   Air yards, snap share, ADOT, deep targets, YAC,
   target separation: NFL tracking/charting data.
   Juke rate, yards created, evaded tackles, explosive
   rating, breakaway runs: PlayerProfiler paid (~$20/mo).
   Contested catch rating: PFF+ paid.

7. A.J. Brown (WR 2019, b2s=15.80) and ~5 others missing
   Wrong link in CFBLink. Needs manual cfb_player_id fix.
```

```
WHAT WOULD IMPROVE THE ANALYSIS

HIGH IMPACT:
  * recruit_rating 2011-2017 (resume CFBD March 2026)
    -> coverage from ~6% to ~100%
    -> could be top-3 predictor for WR

MEDIUM IMPACT:
  * Pre-2016 consensus boards (manual ESPN scrape)
    -> consensus_rank from ~59% to ~100%
  * 2026 NFL combine data (ingest before scoring class)
    -> speed_score for 2026 draft prospects

LOW IMPACT / OPTIONAL:
  * PFF+ YPRR data (~$200/yr)
    -> Particularly improves TE model
  * Manual CFBLink fixes (A.J. Brown + ~5 others)
    -> Restores star players to WR training set
  * PlayerProfiler athleticism tiers (~$20/mo)
    -> Juke rate, yards created, breakaway runs
    -> Could add meaningful signal for RBs
  * PFF+ (~$200/yr) for college YPRR + contested catch
    -> Strong TE signal expected from YPRR

FUTURE MODEL ENHANCEMENTS:
  * Format-specific B2S targets (te_premium_1.0)
  * Penalty-adjusted PPG (fumbles_lost, INTs)
  * NFL aging curves (B2S window extension to year 4-5)
  * Team offensive context features (OC, QB quality)
  * Non-linear model (LightGBM) with monotonic constraints
```

Pos	Feature	N Available	% Missing	Status	Cause
WR	Catch Rate (rec/target)	0	100%	LOW	--
WR	Career Yards Per Target (p	0	100%	LOW	--
WR	Targets (Best Season)	0	100%	LOW	--
WR	Career Yards Per Target (Y	0	100%	LOW	--
WR	Yards Per Target (best sea	0	100%	LOW	--
WR	TD Rate (rec_td/target)	0	100%	LOW	--
RB	Catch Rate (rec/target)	0	100%	LOW	--
RB	Career Yards Per Target (p	0	100%	LOW	--
RB	Targets (Best Season)	0	100%	LOW	--
RB	Career Yards Per Target (Y	0	100%	LOW	--
RB	Yards Per Target (best sea	0	100%	LOW	--
RB	TD Rate (rec_td/target)	0	100%	LOW	--
TE	Career Yards Per Target (Y	0	100%	LOW	--
TE	Catch Rate (rec/target)	0	100%	LOW	--
TE	Targets (Best Season)	0	100%	LOW	--
TE	Yards Per Target (best sea	0	100%	LOW	--
TE	Career Yards Per Target (p	0	100%	LOW	--
TE	TD Rate (rec_td/target)	0	100%	LOW	--